Human-Computer Interaction

7. Mental Model (2)



Hall of Fame or Shame?



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ALPHABETICAL BRAND INDEX # • A • B • C • D • E • F • G • H • I • J • K • L • M • N • O • P • Q • R • S • T • U • V • W • X • Y • Z

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Oh no! We're very sorry, please fix the following issues...

- Please ensure you included a proper phone number.

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NAME (FIRST AND LAST):	James Landay
PHONE NUMBER: (OPTIONAL)	+1 206 651 4382
EMAIL ADDRESS:	landay@gmail.com
CONFIRM EMAIL ADDRESS:	landay@gmail.com

(•) Contact me by email
( ) Contact me by phone

REGARDING:	Returns
ORDER NUMBER: (OPTIONAL)	119342675
MESSAGE SUBJECT LINE:	didn't get credited for 1 pair returned

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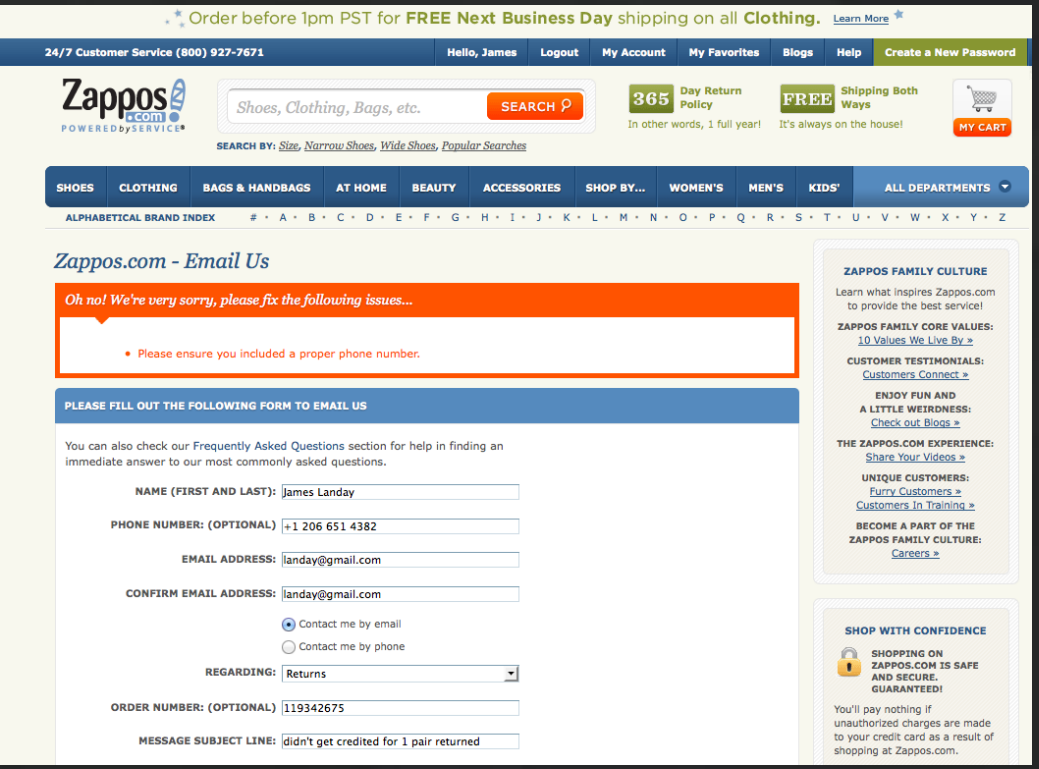
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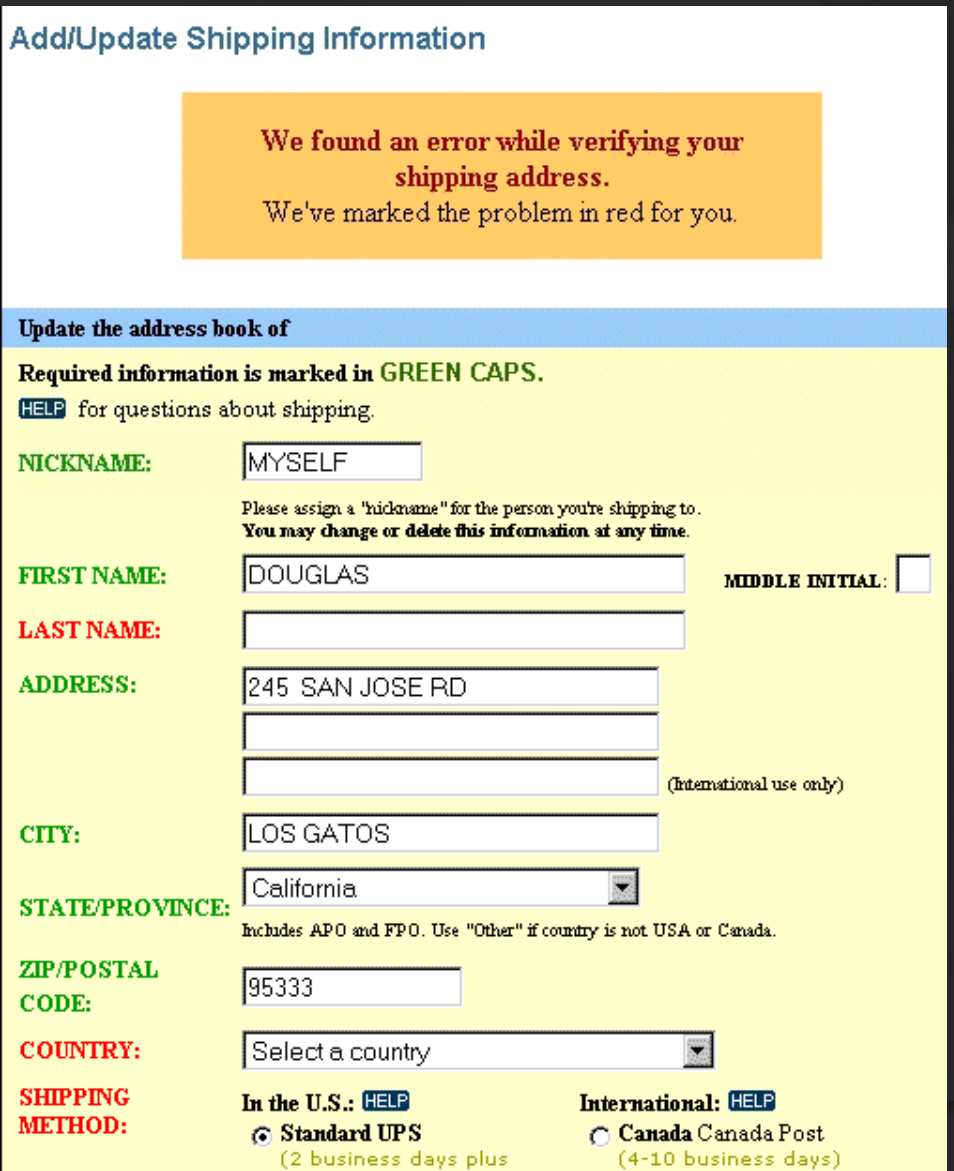
Hall of Shame!



- Error Messages
  - Where is error?
  - What's wrong with it?
  - Parse it & fix it yourself!

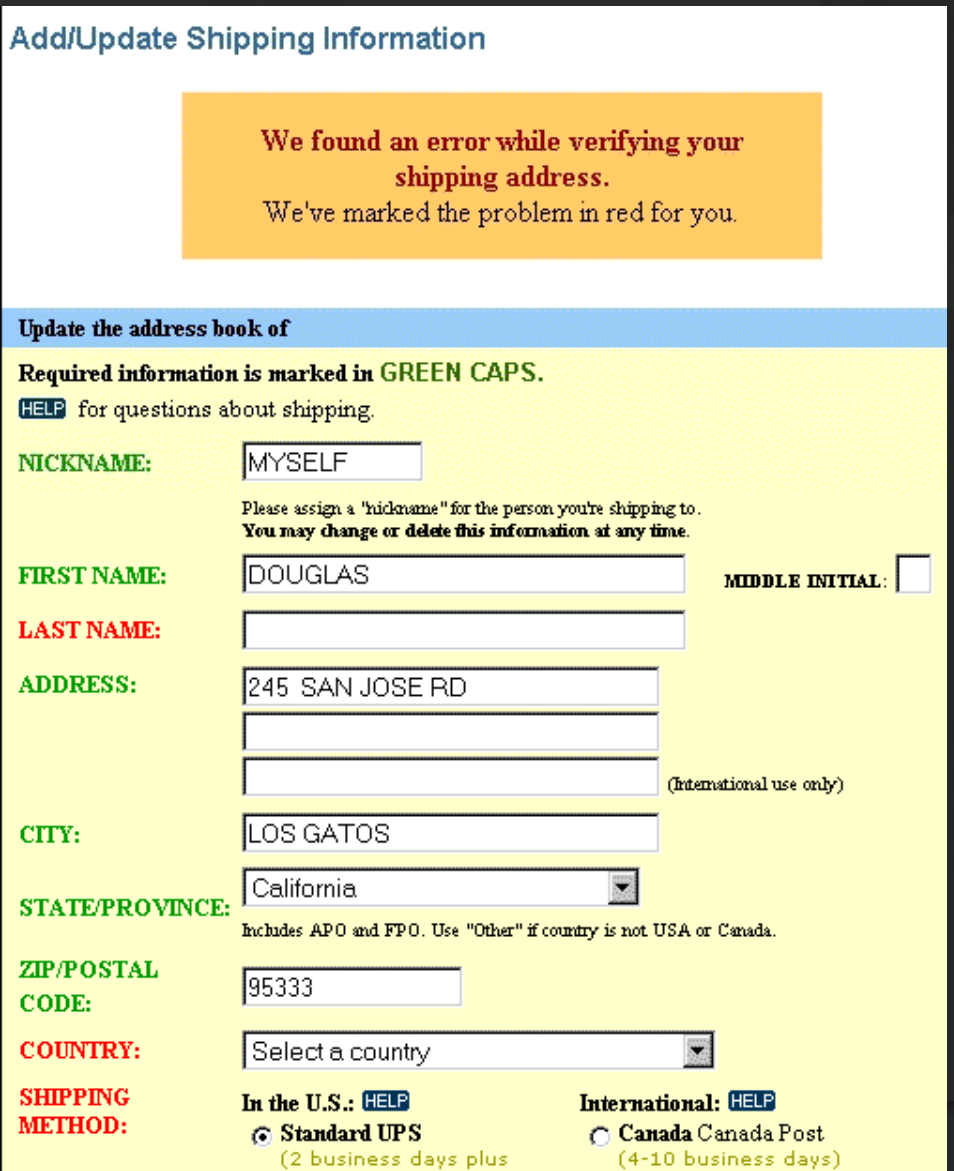
Hall of Fame or Shame?





- Design based on a top retailer's site
- In study, user could not get by this screen, why?

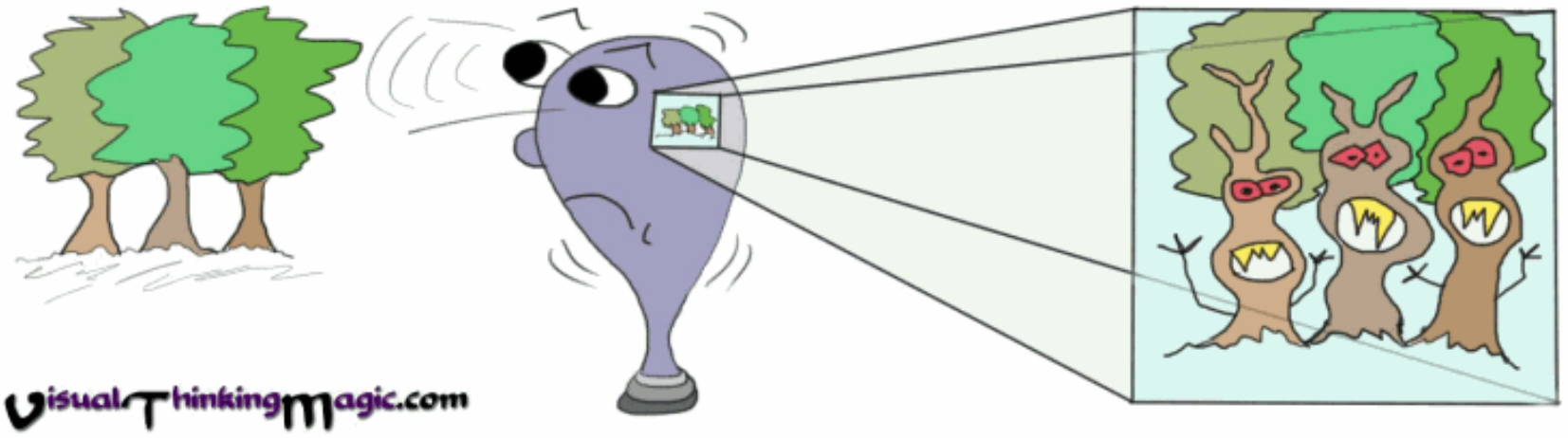
Hall of Shame!



- Design based on a top retailer's site
- In study, user could not get by this screen, why?
- Color deficiency
  - can't distinguish between red & green
- How to fix?
  - redundant cues

Recap: Mental model

"How your mind thinks that something works"

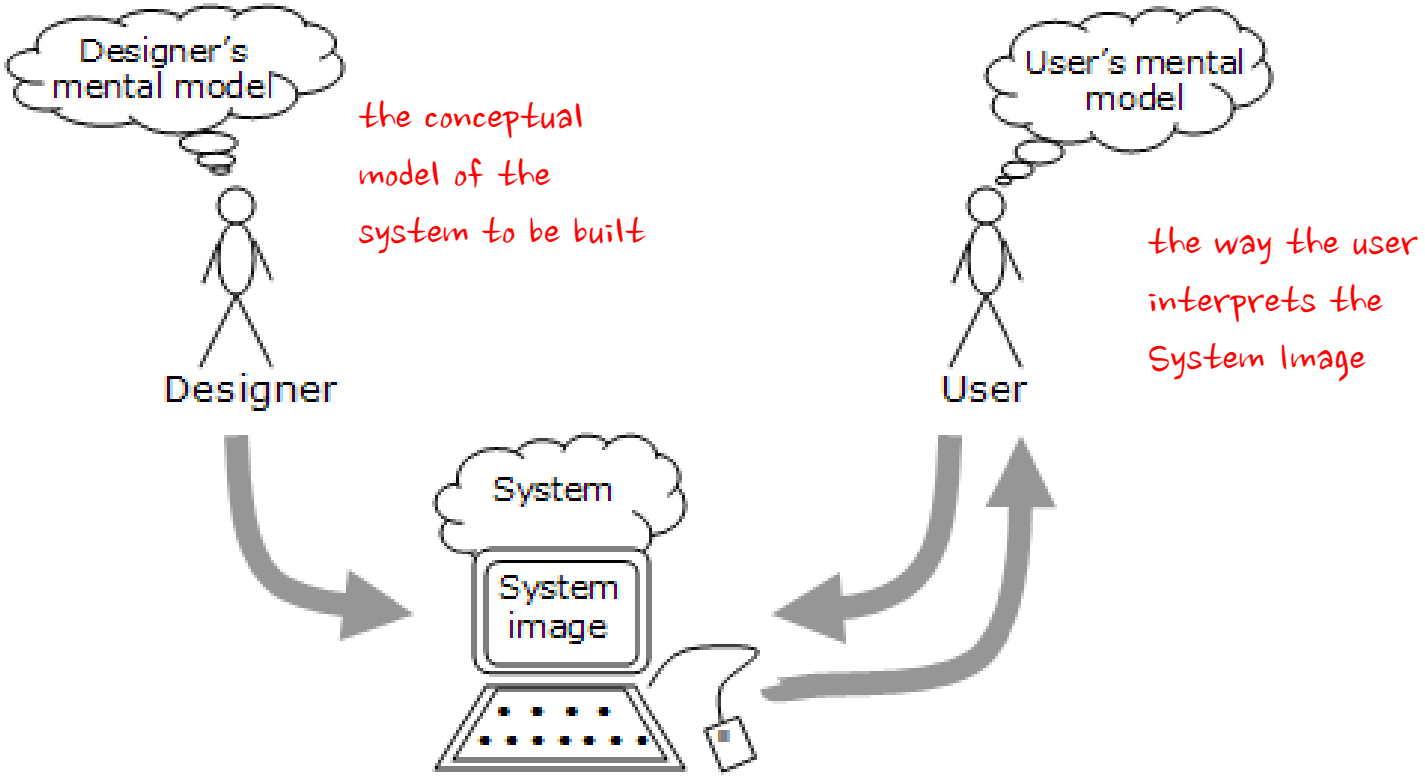


Recap: Mental model

"The models people have of themselves, others, the environment and the things with which they interact. People form mental models through experience, training and instruction"

- An explanation of someone's thought process about how something works in the real world
- Enable people to reason about a system
- Affect the way we see and interpret reality
- When users approach an unfamiliar system, they subconsciously refer to their mental model, User's model

Recap: User's vs. designer's model

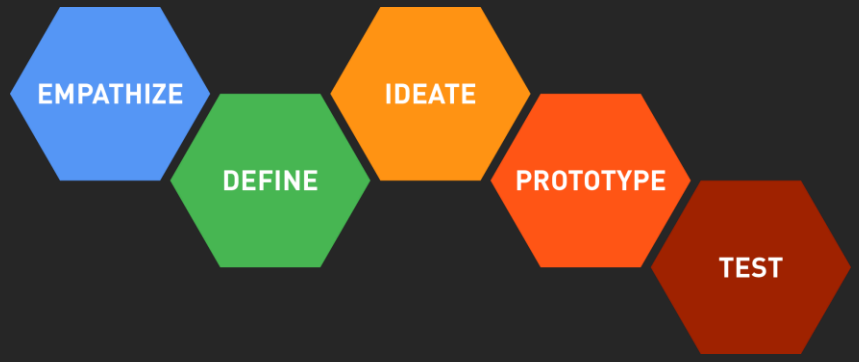


As a designer, you want to use the system's UI to accurately convey your mental model of the system to the user.

Today's agenda

Model human processor

- Perceptual processor
- Working memory
- Cognitive processor
- Motor processor



Model Human Processor

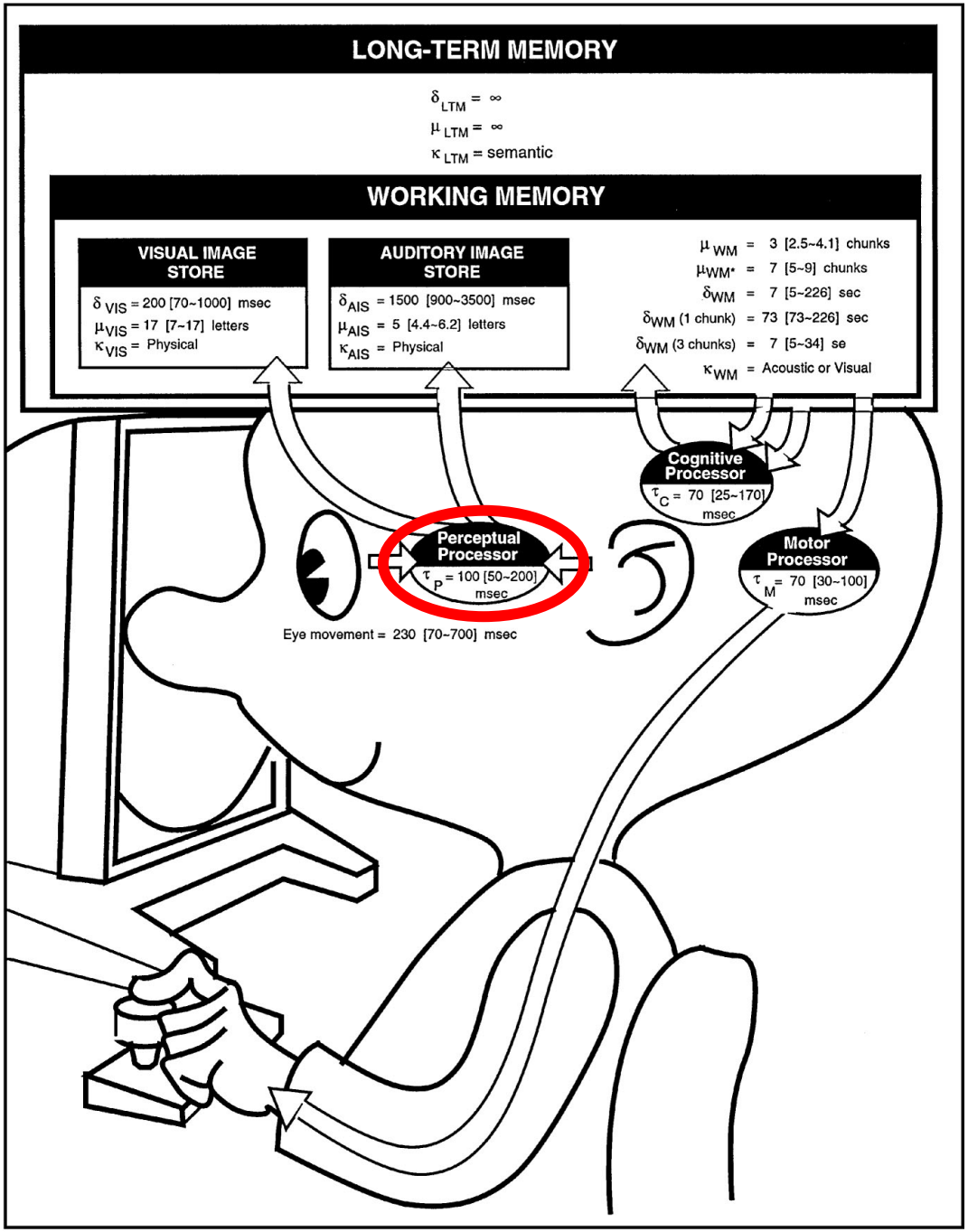
The Model Human Processor

- Published by Card, Moran & Newell, 1983, in the book "The Psychology of Human-Computer Interaction"

- Calculate how long it takes to perform a certain task
- Consider humans as information processing systems
- Consist of three interacting processors
    1. Perceptual processor: Responsible for transforming external environment into a form that cognitive system can process
    2. Cognitive processor: Uses contents of working memory and long-term memory to make decisions and schedule actions with motor system
    3. motor processor: Translates thoughts into actions

Why is MHP Useful?

- Use model to:
  - Predict and compare usability of different interface designs
    - Task performance, learnability, and error rates
    - No users or functional prototype required!
  - Apply model to predict effectiveness of new interfaces
  - Develop guidelines for interface design: Color, spatial layout, recall, response rates, etc.


LONG-TERM MEMORY
$\delta_{LTM} = \infty$
$\mu_{LTM} = \infty$
κ_{LTM} = semantic
WORKING MEMORY
VISUAL IMAGE STORE
δ_{VIS} = 200 [70~1000] msec
μ_{VIS} = 17 [7~17] letters
κ_{VIS} = Physical
AUDITORY IMAGE STORE
δ_{AIS} = 1500 [900~3500] msec
μ_{AIS} = 5 [4.4~6.2] letters
κ_{AIS} = Physical
μ_{WM} = 3 [2.5~4.1] chunks
μ_{WM^*} = 7 [5~9] chunks
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δ_{WM} (1 chunk) = 73 [73~226] sec
δ_{WM} (3 chunks) = 7 [5~34] se
κ_{WM} = Acoustic or Visual
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τ_C = 70 [25~170] msec
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τ_P = 100 [50~200] msec
Motor Processor
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Eye movement = 230 [70~700] msec

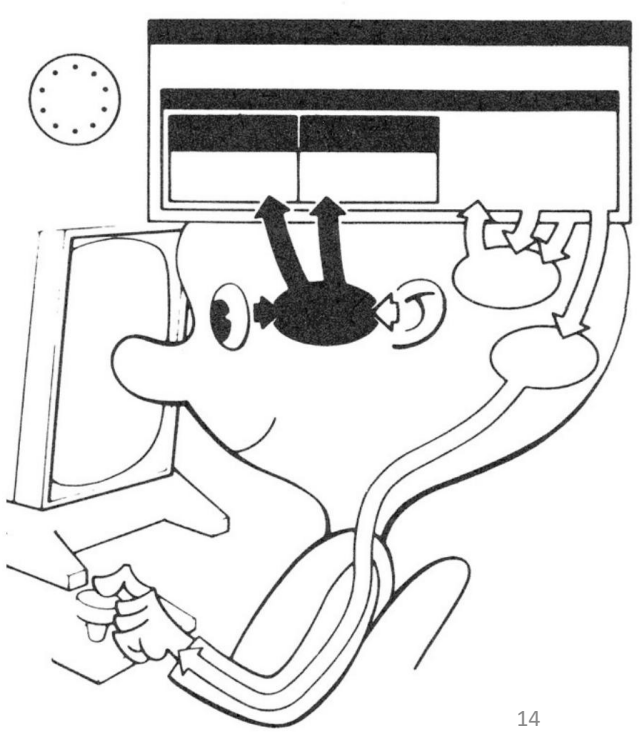
(1) Perceptual processor

Perceptual processor is responsible for transforming external environment into a form that cognitive system can process.

Physical store from our senses: sight, sound, touch,…

- Code directly based on sense used
- Visual, audio, haptic, … features

Decay time for working memory: 200ms



Pre-attentive Processing

Preattentive: a limited set of visual properties that are detected very rapidly and accurately by the low-level visual system without our conscious action.

- All available information is pre-attentively processed. Then, the brain filters and processes what is important.
- It takes less than 500 milliseconds for the eye and the brain to process a preattentive property of any image.
- These properties can be harnessed to make it easier for a user to understand what is presented through the design unconsciously.
- It allows the designer to grab the attention of the user without their input or effort.

https://www.youtube.com/watch?v=09jy3Znji74

Pre-attentive Processing

Three preattentive visual properties as follows:

- Color
- Form
- Movement

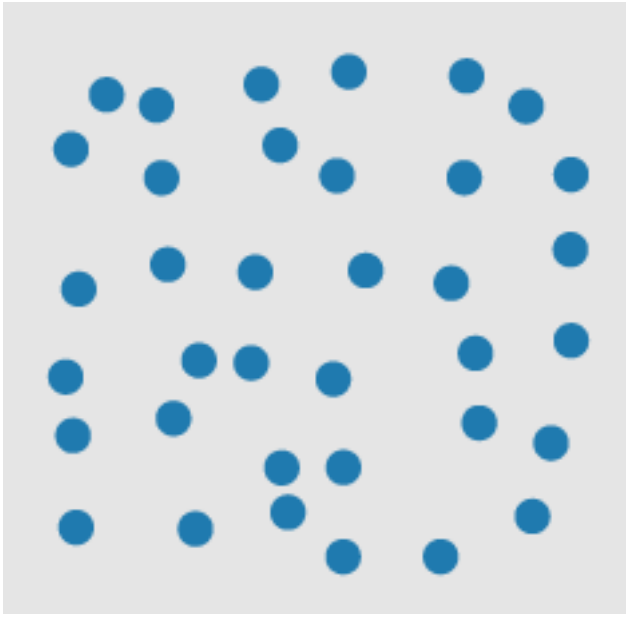
How many 3’s?

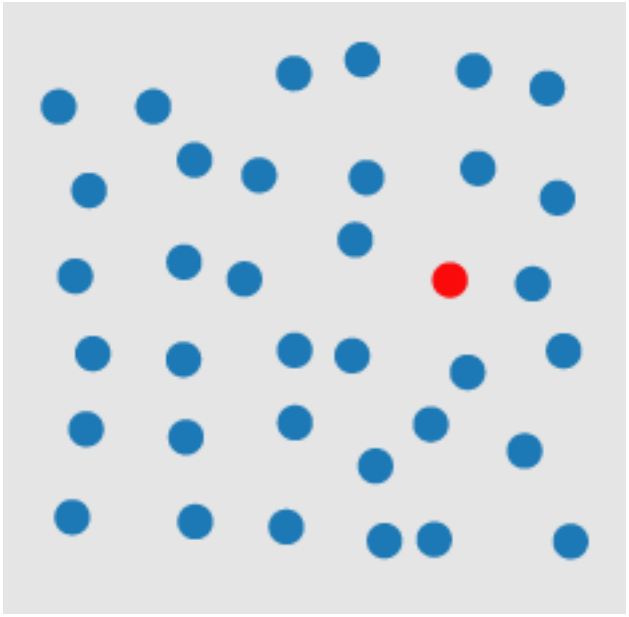
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How many 3’s?

128176875613897654698450698560498282676 2
980985845822450985645894509845098094 3585
909103020990595959577256467505067890456 7
884578980982167765487636490856091294968 6

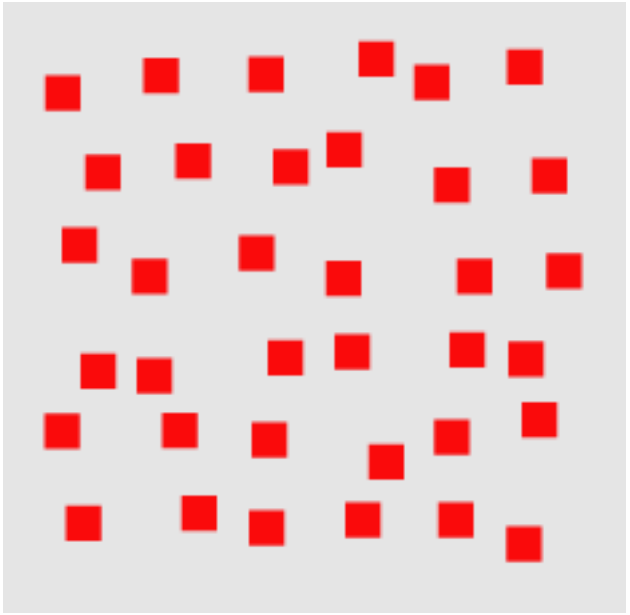
Visual pop-out: color (hue)

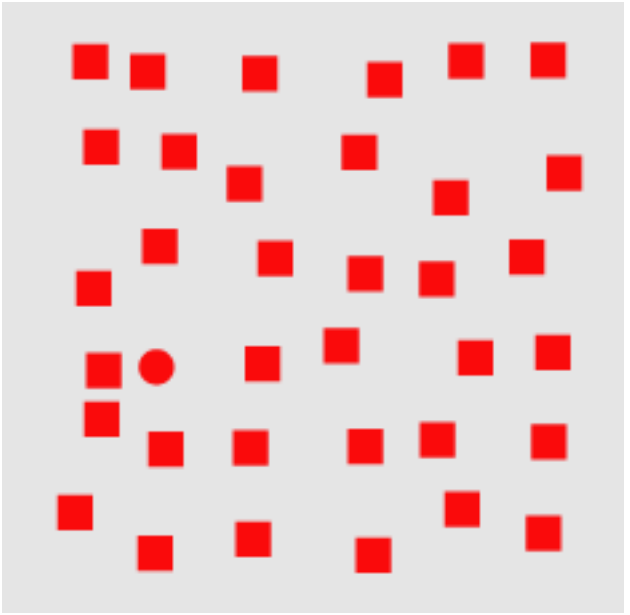




Resources from : https://www.csc2.ncsu.edu/faculty/healey/PP/index.html

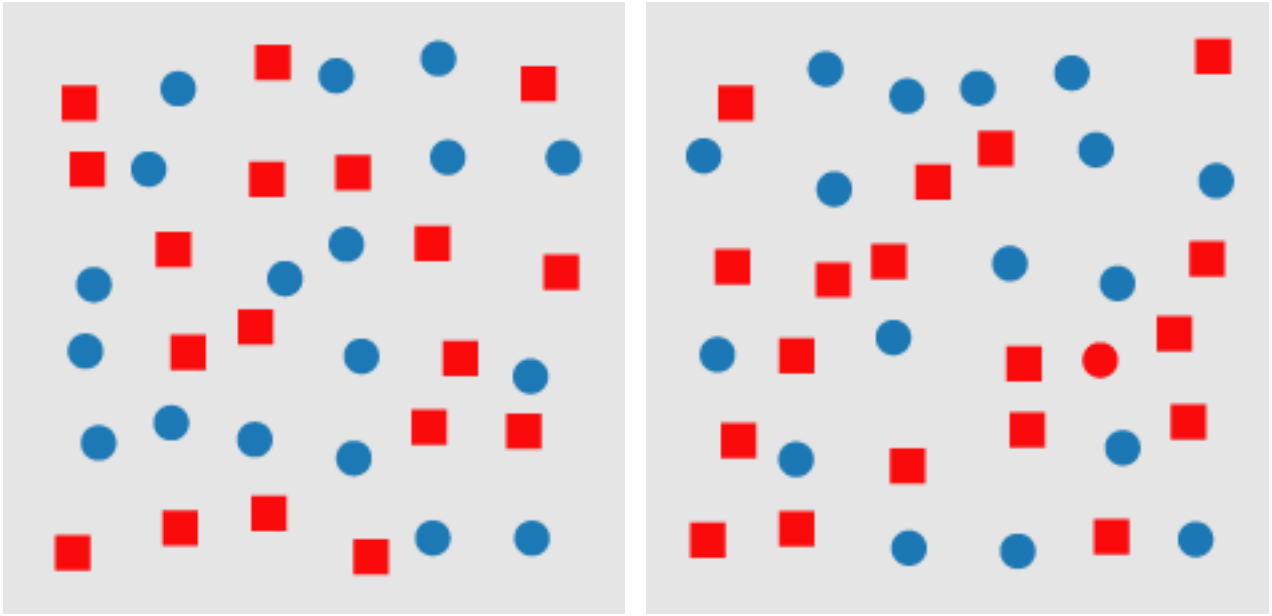
Visual pop-out: shape (curvature)





Resources from : https://www.csc2.ncsu.edu/faculty/healey/PP/index.html

Feature conjunctions



Resources from : https://www.csc2.ncsu.edu/faculty/healey/PP/index.html

Change blindness



Change blindness

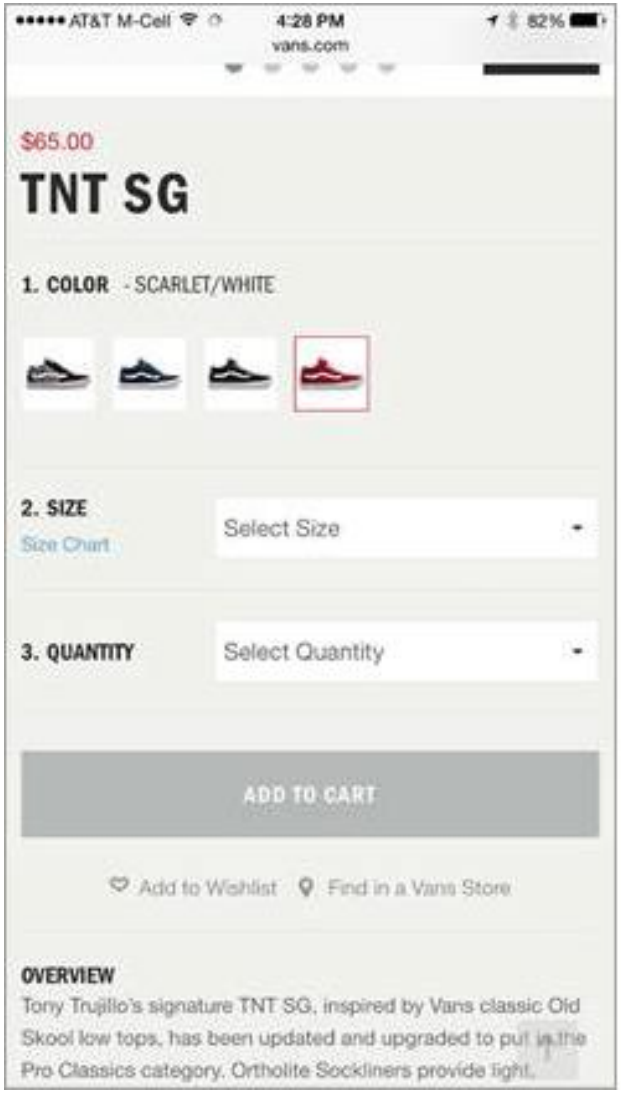


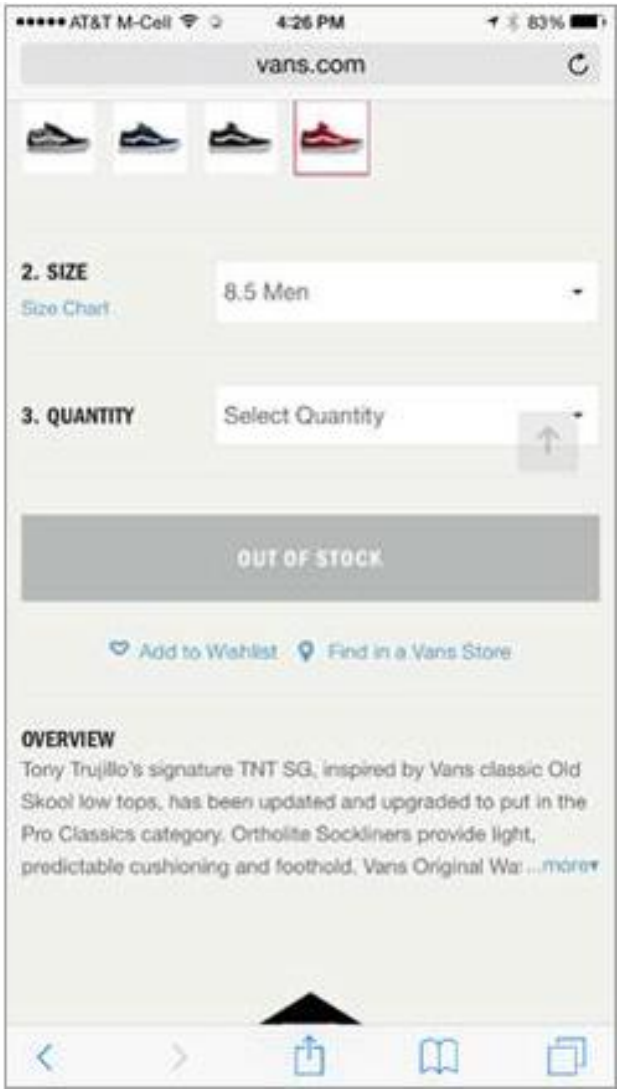
Change blindness

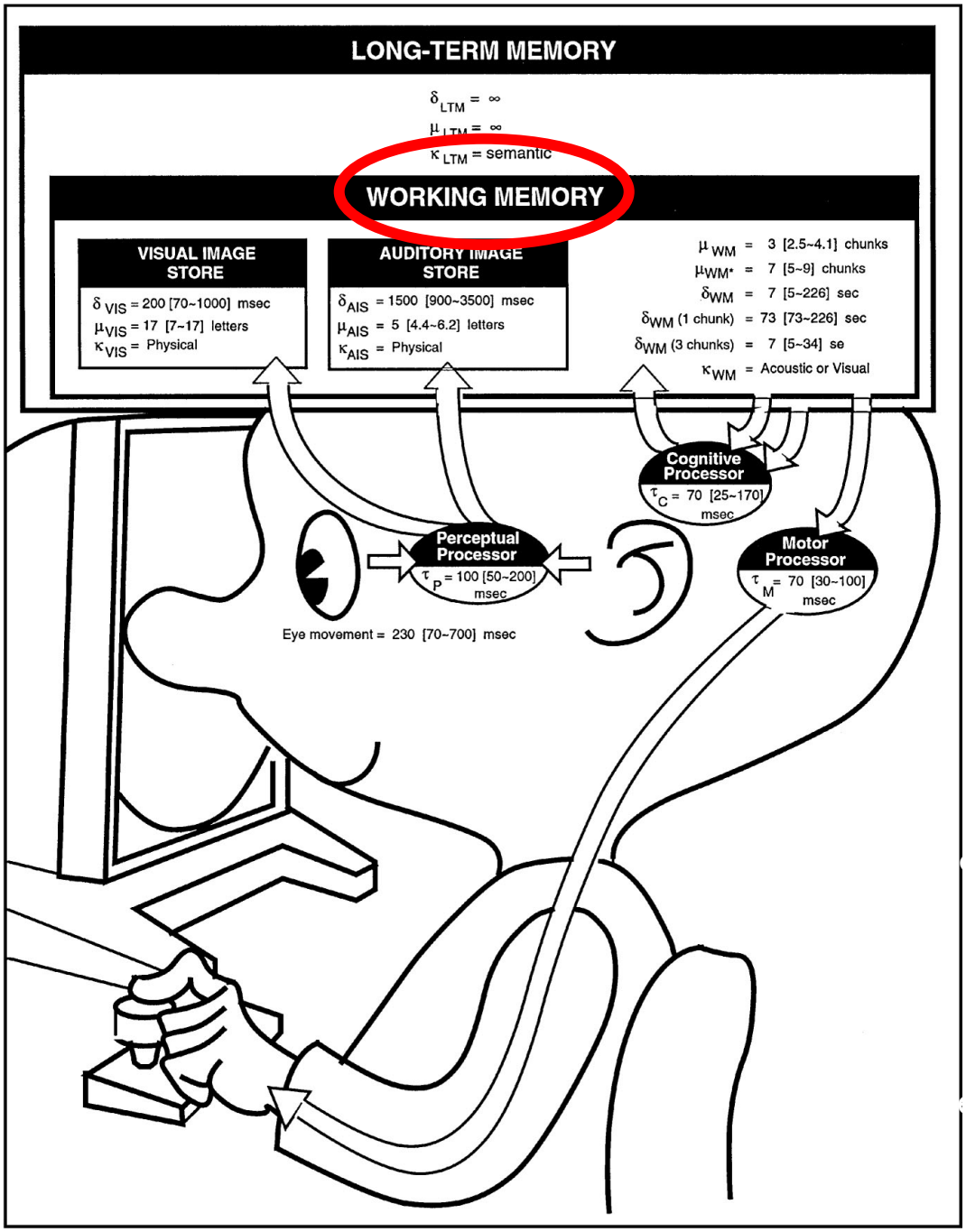
A perceptual phenomenon that occurs when a change in a visual stimulus is introduced and the observer does not notice it

- Our visual intelligence can only detect changes in those parts of the image to which we explicitly attend.
- People often overlook new visual details added to an existing image. This change blindness can affect critical information such as error messages and navigation menus, leading to user confusion and task failure. Luckily, with the right visual presentation you can dramatically reduce the likelihood of change blindness.

Change blindness






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Working memory

Working memory is small

- Temporary storage
- Decay in 200ms

Working memory holds intermediate products of thinking and representations produced by perceptual system

- Comprised of activated sections of LTM called “chunks”
- A chunk is a hierarchical symbol structure
- 7 ± 2 chunks active at any given time

Working memory / Short Term Memory

35 x 6

- A scratch-pad for temporary recall of information
- Accessed rapidly: about 70ms
- Decays rapidly: about 200ms

Short Term Memory

Two measures for memory capacity

- Determine the length of a sequence which can be remembered in order
- Freely recalled in any order

4048942000

Short Term Memory

404-894-2000

About Chunks

- A chunk is a meaningful grouping of information – allows assistance from LTM
- 4793619049 vs. 704 687 8376
- NSAFBICIANASA vs. NSA FBI CIA NASA
- My chunk may not be your chunk
  - User and task dependent

Short Term Memory

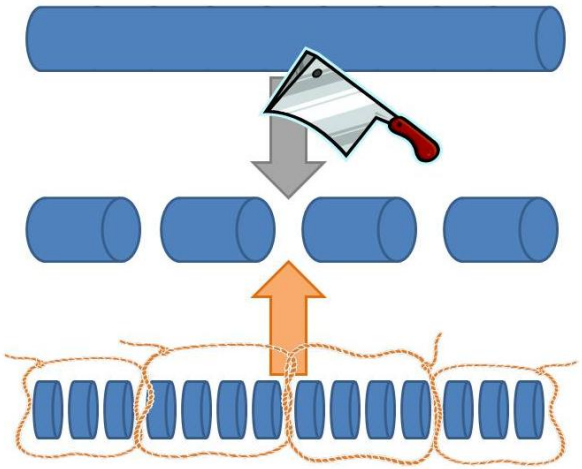
- Use “chunks”: 7 +- 2 units of information
- Symbolic, nonphysical acoustic or visual coding
- Decay 5-226 sec, rehearsal prevents decay
- Another task prevents rehearsal - interference

The Magic number 7 ± 2

- George Miller's theory of how much information people can remember
- People's immediate memory capacity is very limited
- The number of objects an average human can hold in working memory (short-term) is 7 ± 2
- "Chunking": breaking up long strings of information into units

Chunking

2 Views of Chunking





Products Often Bought With This:



Mobile App or Game Reskinning
$750
Add to cart



Mobile Application QA and Testing
$1,999
Add to cart



Project Specification Workshop
$1,499
Add to cart



Mobile App Landing Page Design
$799
Add to cart

Chunking





2038675309

203-867-5309

05082015

05/08/2015

Chunking

Hierarchy

- Fish
  - BMWs
  - Doors
  - Numbers
  - Friends
- Armchairs
  - People
  - Drums
  - Computers
  - Singers
- Monkeys
  - Beers

- Fruits
- Lemons
- Apples
- Pears
- Grapes
- Vegetables
- Cucumbers
- Lettuce
- Celery
- Carrots
- Meats
- Mutton

When to Use Chunking in UI

The primary purpose of chunking is the enhancement of working memory.

- Ideal when specific information must be memorized for later use
- Ideal in environments where an interface must compete against other stimuli for the attention or working memory of the end user (car navigation systems, cell phone, public kiosks)
- Effective use of chunking can improve the usability and effectiveness of an information system.

When Not to Use Chunking in UI

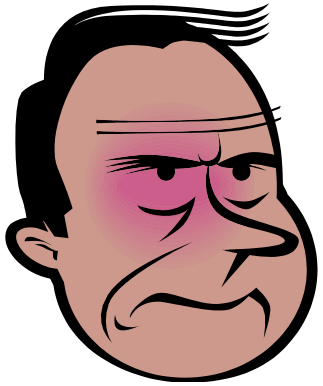
The primary purpose of chunking is the enhancement of working memory.

- Should not be used when the information must be searched, scanned, or analyzed
  - Search engine results are an example of information that does not need to be memorized
- Should not be used as an argument for improved simplicity, legibility, or uncluttered page design

Long-Term Memory

- Seemingly permanent & unlimited
- Access is harder, slower
  - -> Activity helps (we have a cache)

File system full



- Retrieval depends on network of associations
- How information is perceived, understood and encoded determines likelihood of retrieval

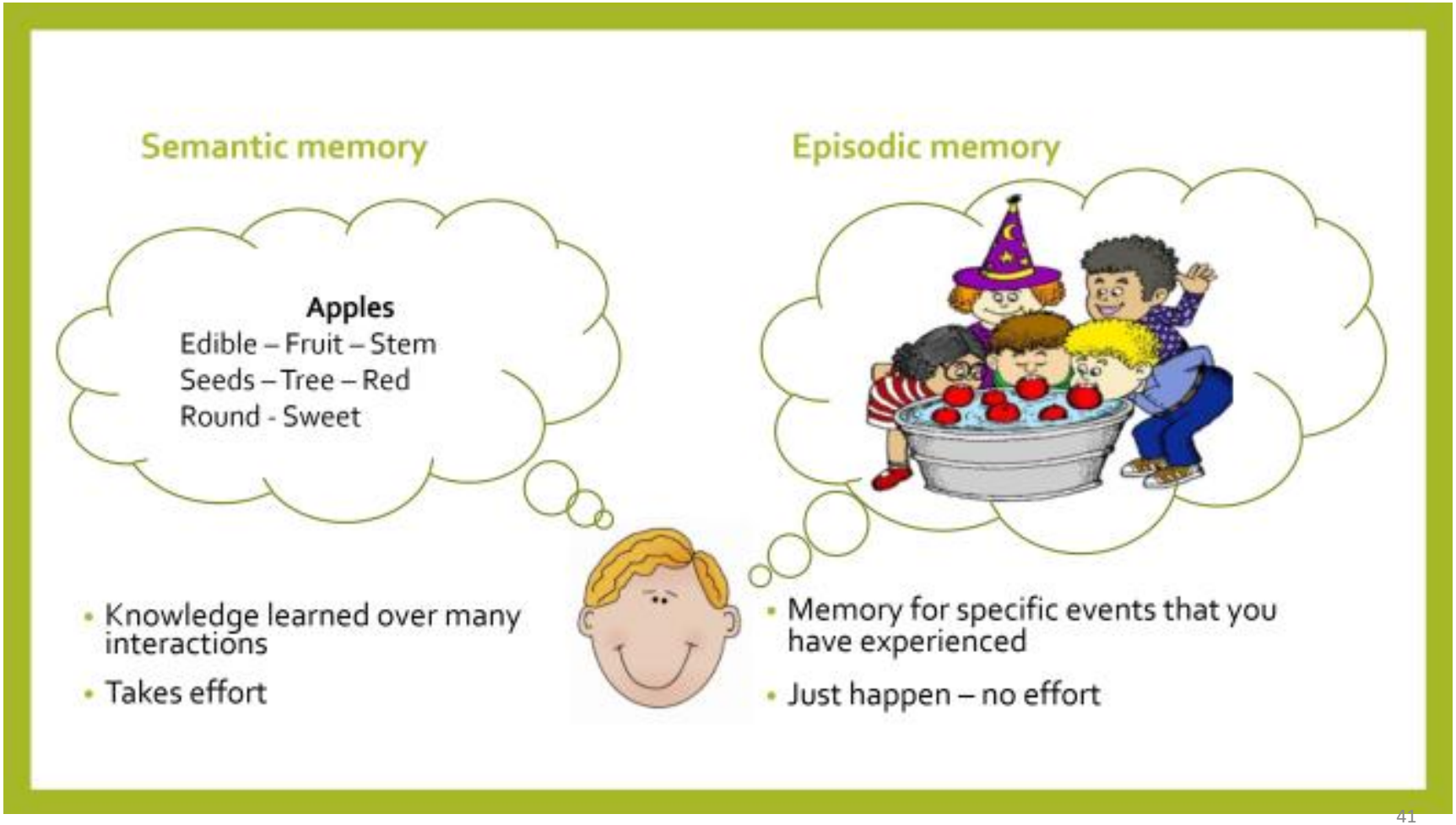
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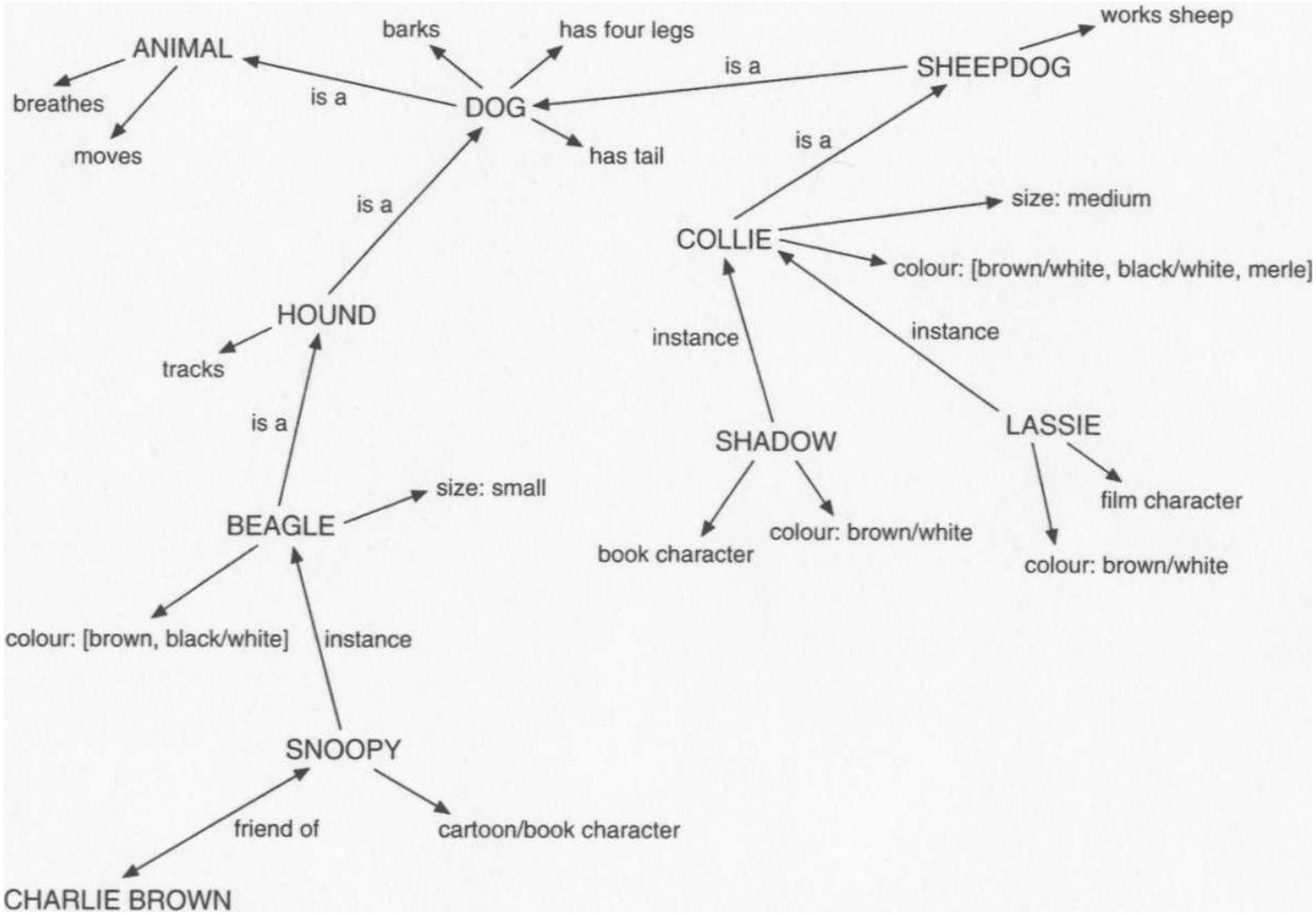
- Episodic memory
  - Events & experiences in serial form
    - Helps us recall what occurred

- Semantic memory
  - Structured record of facts, concepts & skills
    - One theory says it’s like a network
    - Another uses frames & scripts

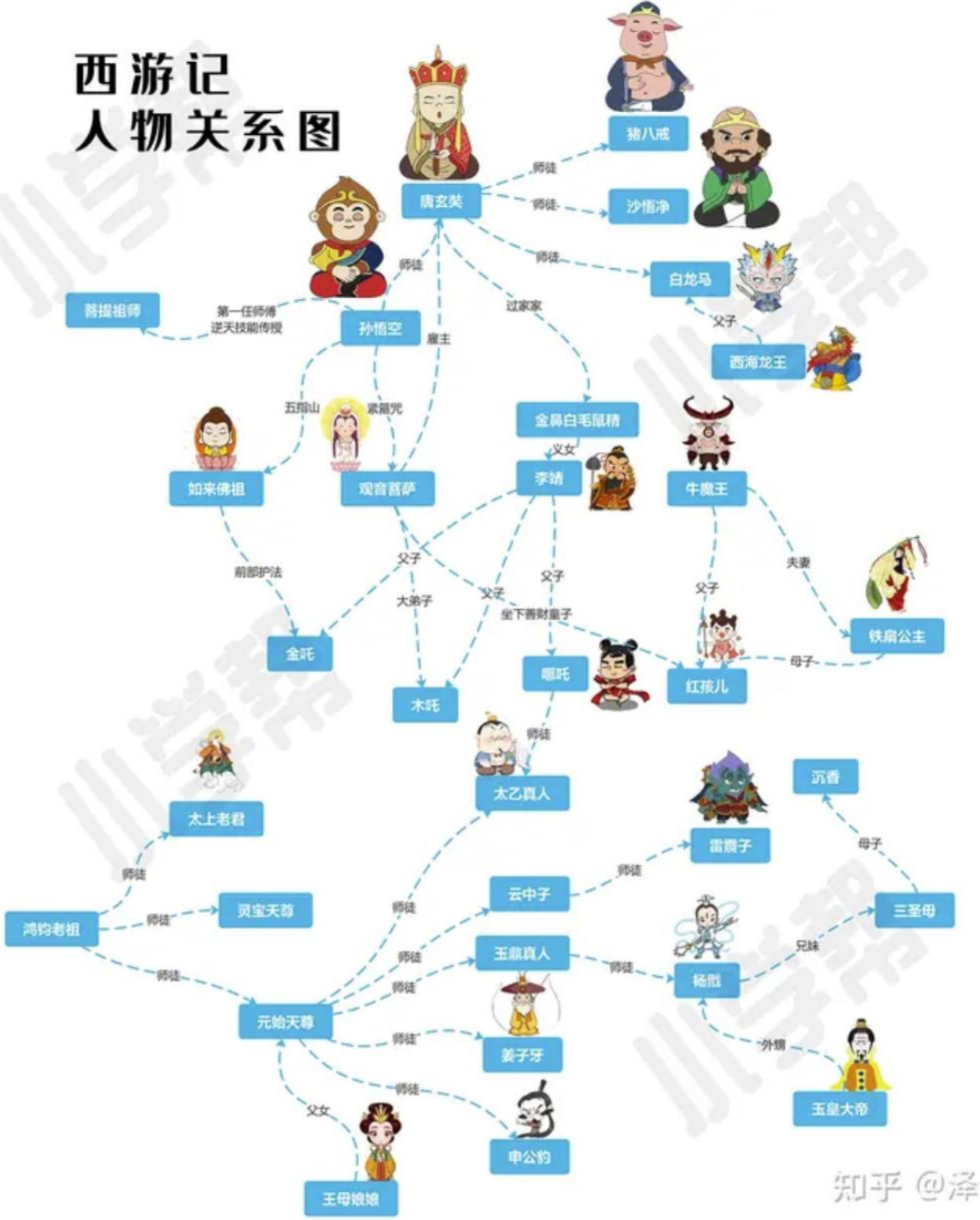
LT Memory Structure



LT Memory Structure



https://academic-accelerator.com/encyclopedia/zh-cn/semantic-memory



https://academic-accelerator.com/encyclopedia/zh-cn/semantic-memory

LT Memory Processes

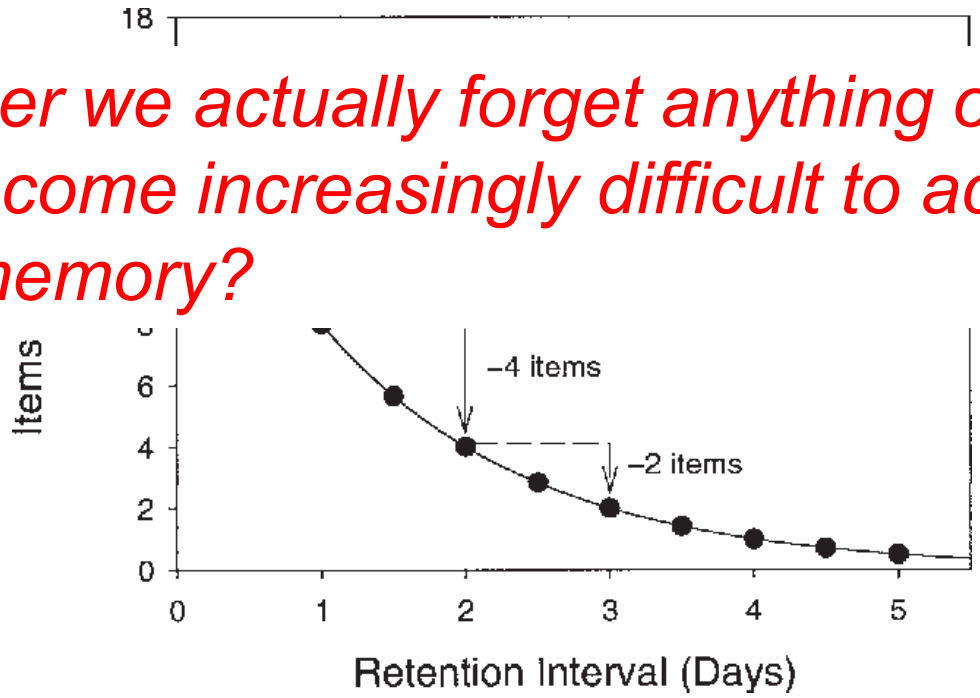
- Store / Remembering information:
  Things move from STM to LTM by rehearsal & practice and by use in context
  - Total time hypothesis
  - Distribution of practice effect
  - Visualize, Sentences

Faith Age Cold Tenet Quiet Logic Idea Value

Boat Tree Cat Child Rug Plate Church Gun

LT Memory Processes

- Forgetting
  We *forget* things due to *decay* and *interference*
  - Similar gets in the way



Whether we actually forget anything or whether it just become increasingly difficult to access items from memory?

LT Memory Processes

- Information Retrieval
  We **recognize** information easier than we can **recall** information

- Examples?
- Implications?

Recognition over recall

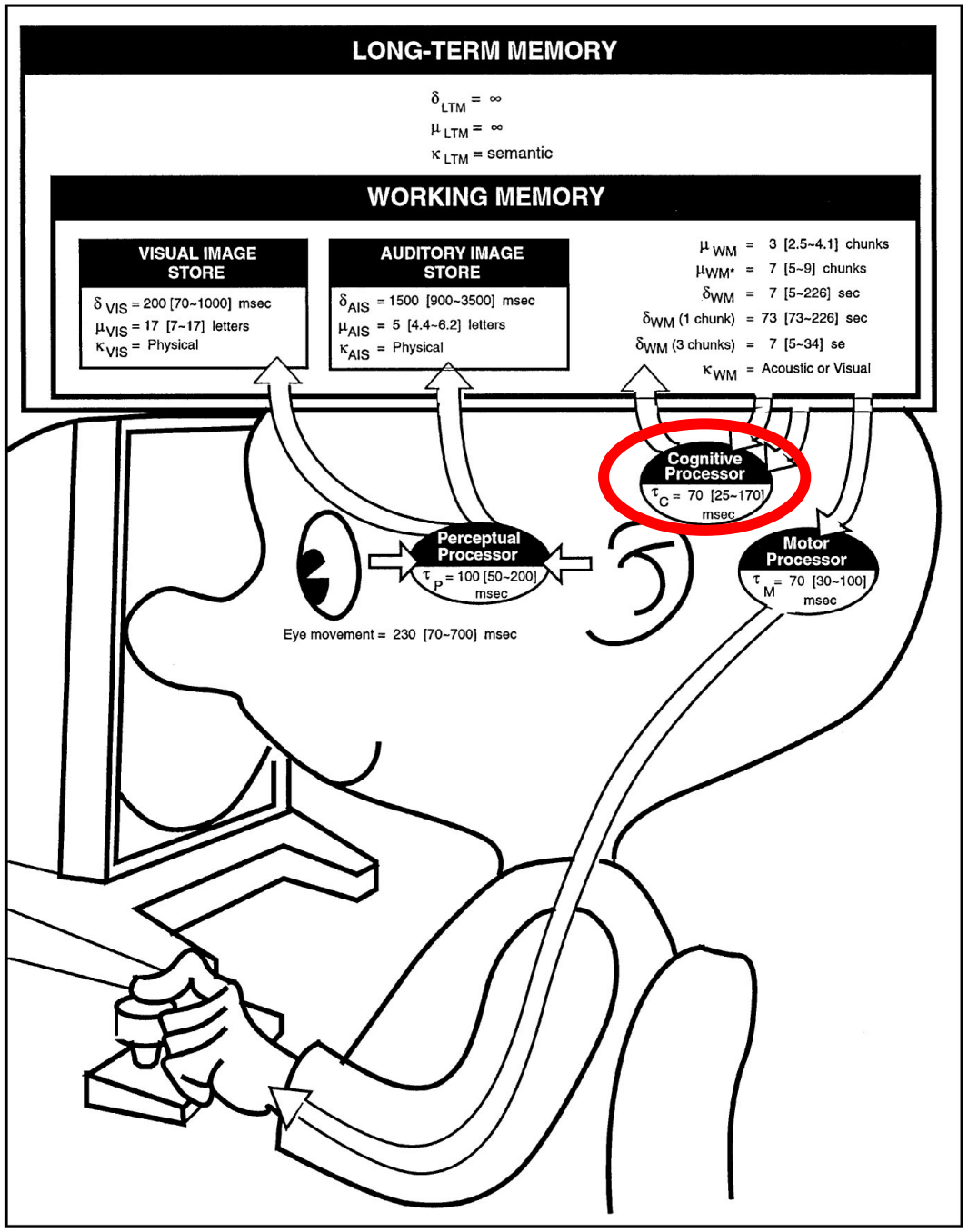
- Recall – information reproduced from memory
- Recognition
  - Presentation of information provides cues that information has been encountered before
  - Easier because of cues (context) for retrieval

Recognition over recall

- Command line interface (recall) vs. GUI (recognition) interface

```
[root@localhost ~]# ping -q fa.wikipedia.org
PING text.pmtpa.wikimedia.org (208.80.152.2) 56(84) bytes of data.
^C
--- text.pmtpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 540.528/540.528/540.528/0.000 ms
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x.  2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x.  3 root root 4096 May 18 16:03 db
drwxr-xr-x.  3 root root 4096 May 18 16:03 empty
drwxr-xr-x.  2 root root 4096 May 18 16:03 games
drwxrwx--T.  2 root gdm  4096 Jun  2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x.  2 root root 4096 May 18 16:03 local
lrwxrwxrwx.  1 root root   11 May 14 00:12 lock -> ../run/lock
drwxr-xr-x. 14 root root 4096 Sep 14 20:42 log
lrwxrwxrwx.  1 root root   10 Jul 30 22:43 mail -> spool/mail
drwxr-xr-x.  2 root root 4096 May 18 16:03 nis
drwxr-xr-x.  2 root root 4096 May 18 16:03 opt
drwxr-xr-x.  2 root root 4096 May 18 16:03 preserve
drwxr-xr-x.  2 root root 4096 Jul  1 22:11 report
lrwxrwxrwx.  1 root root    6 May 14 00:12 run -> ../run
drwxr-xr-x. 14 root root 4096 May 18 16:03 spool
drwxrwxrwt.  4 root root 4096 Sep 12 23:50 tmp
drwxr-xr-x.  2 root root 4096 May 18 16:03 yp
[root@localhost var]# yum search wiki
Loaded plugins: langpacks, presto, refresh-packagekit, remove-with-leaves
rpmfusion-free-updates                                          | 2.7 kB     00:00
rpmfusion-free-updates/primary_db                               | 206 kB     00:04
rpmfusion-nonfree-updates                                       | 2.7 kB     00:00
updates/metalink                                                | 5.9 kB     00:00
updates                                                         | 4.7 kB     00:00
updates/primary_db                    73% [=================       ]  62 kB/s | 2.6 MB     00:15 ETA
```




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(2) Cognitive processor

Cognitive processor uses contents of working memory and long-term memory to make decisions and schedule actions with motor system

Typical matching time

- Digits: 33ms
- Colors: 38ms
- Geometry: 50ms…

Decision time increases with the uncertainty about the judgment to be made, which requires more cognitive cycles.

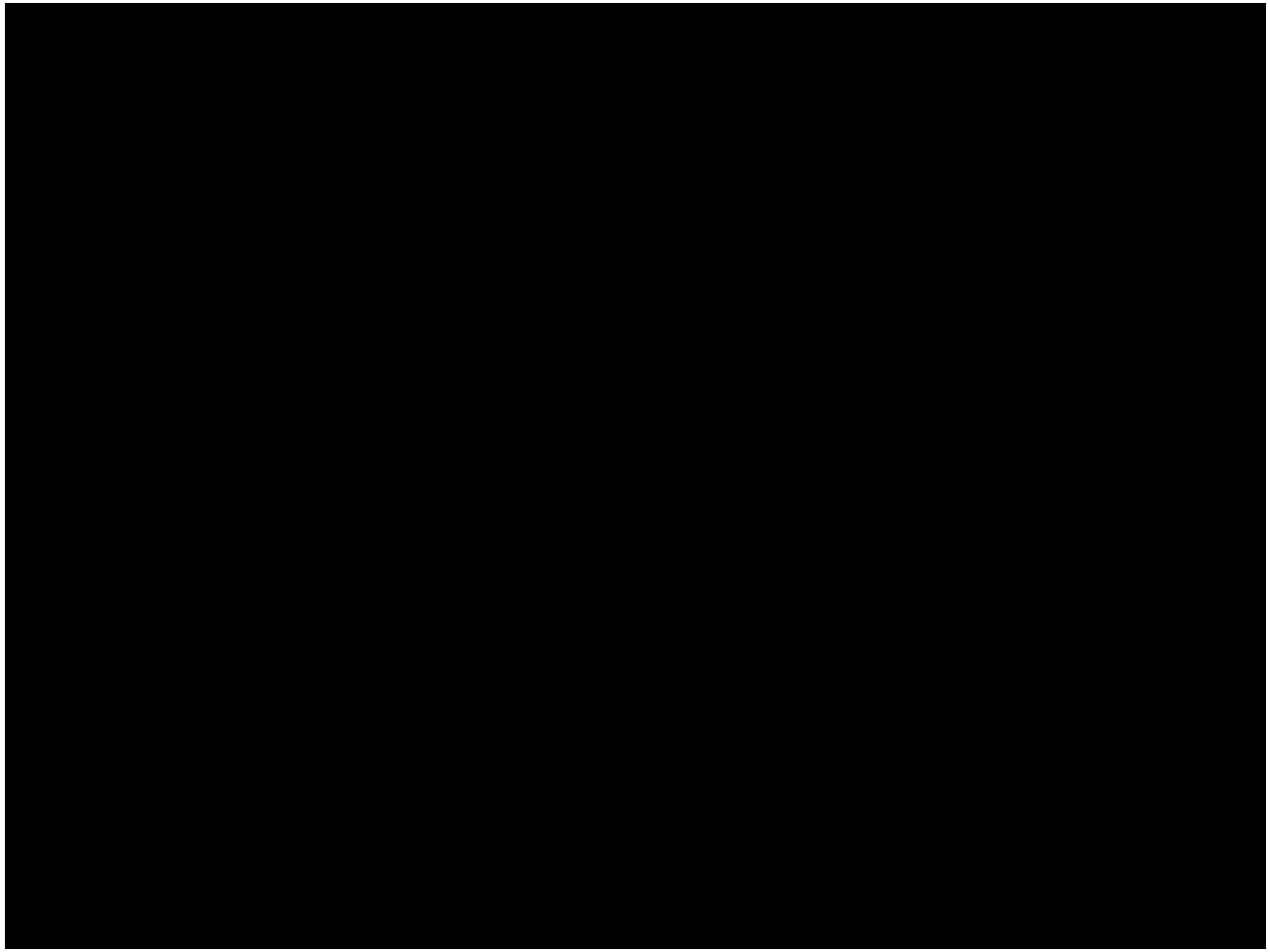
Selective Attention

- We can focus on one particular thing
  - Cocktail party chit-chat

- Salient visual/auditory cues facilitate s.a.
  - Examples?
    - Boldface, blinking and beeping

- Visual or Auditory Streams form after a few seconds

Selective attention



https://www.youtube.com/watch?v=vJG698U2Mvo

Selective attention (Tunnel Vision)

"Users don't see stuff that's right on the screen. Selective attention makes people overlook things outside their focus of interest."

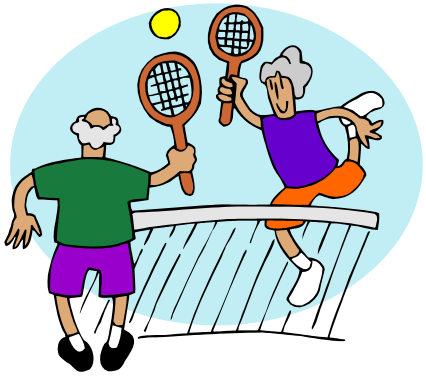
- The act of focusing on a particular object for a period of time while simultaneously ignoring irrelevant information that is also occurring
- This occurs on a daily basis and can be seen in basically any of your interactions. Because it is impossible to give attention to every stimulus in our environment, we use selective attention to select what stimuli are important as events occur.

Selective Attention



Learning

- Two types:
  - Procedural – How to do something
  - Declarative – Facts about something
- Involves
  - Understanding concepts & rules
  - Memorization
  - Acquiring motor skills
  - Automotization
    - Tennis
    - Driving to work
      - Even when don't want to
    - Swimming, Bike riding, Typing, Writing



Learning

- Facilitated
  - By structure & organization
  - By similar knowledge, as in consistency in UI design
  - By analogy
  - If presented in incremental units
  - Repetition
- Hindered
  - By previous knowledge
    - Try moving from Mac to Windows

=> Consider user's previous knowledge in your interface design

Problem Solving

- Storage in LTM, then application
- Reasoning
  - Deductive - If A, then B
  - Inductive - Generalizing from previous cases to learn about new ones
  - Abductive - Reasons from a fact to the action or state that caused it
- Goal in UI design - facilitate problem solving!
  - How??

Observations (for UI design)

- Users focus on getting job done, not learning to effectively use system
- Users apply analogy even when it doesn't apply
  - Or extend it too far - which is a design problem
    - Dragging floppy disk icon to Mac's trash can does NOT erase the disk, it ejects disk!

Observations (for UI design)

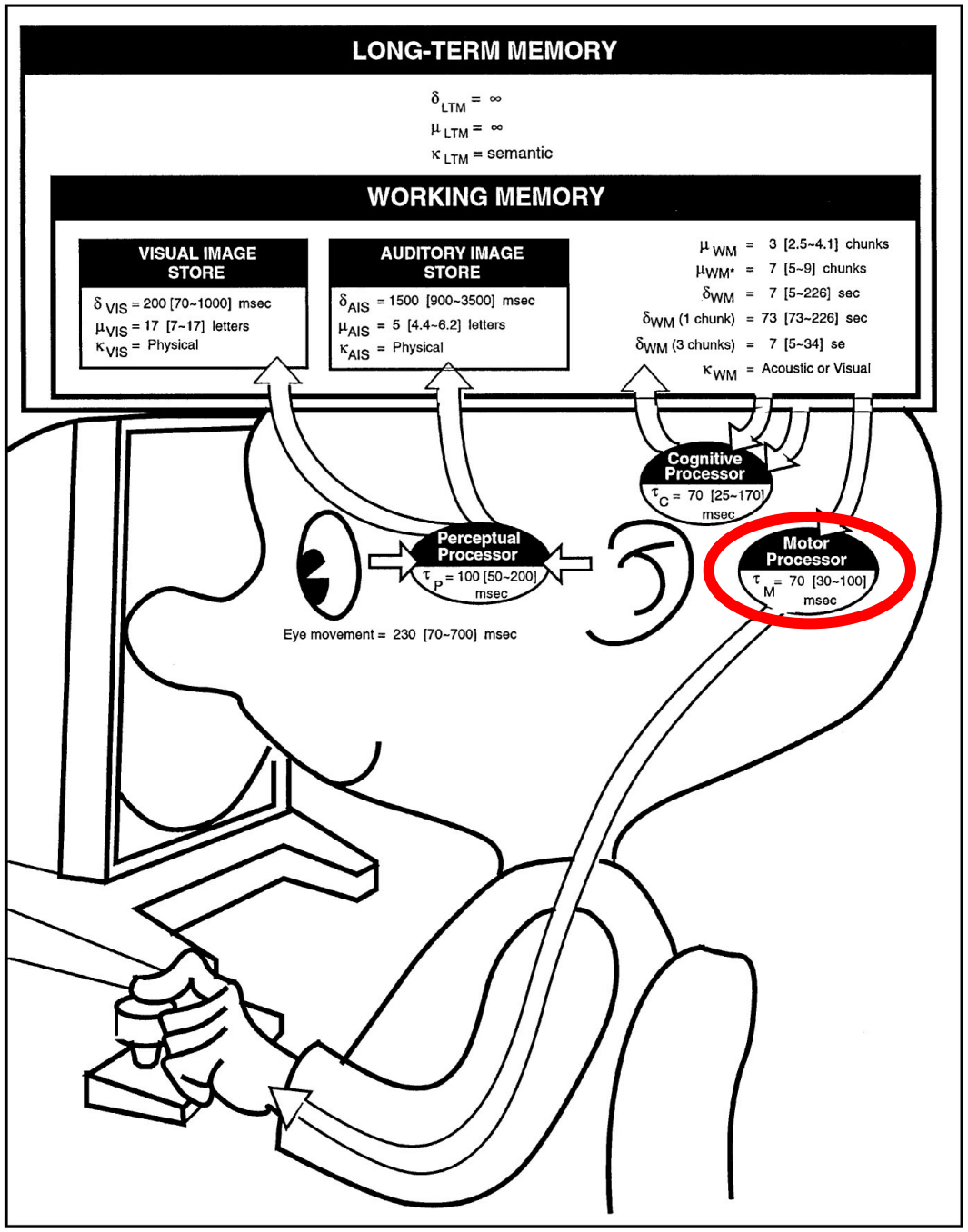
- We are more heuristic than algorithmic
  - We try a few quick shots rather than plan
    - Resources simply not available
- We often choose suboptimal strategies for low priority problems
- We learn better strategies with practice

Implications (for UI design)

- Allow flexible shortcuts
  - Forcing plans will bore user



- Have active rather than passive help
  - Recognize waste


LONG-TERM MEMORY
$\delta_{LTM} = \infty$
$\mu_{LTM} = \infty$
κ_{LTM} = semantic
WORKING MEMORY
VISUAL IMAGE STORE
δ_{VIS} = 200 [70~1000] msec
μ_{VIS} = 17 [7~17] letters
κ_{VIS} = Physical
AUDITORY IMAGE STORE
δ_{AIS} = 1500 [900~3500] msec
μ_{AIS} = 5 [4.4~6.2] letters
κ_{AIS} = Physical
μ_{WM} = 3 [2.5~4.1] chunks
$\mu_{WM'}$ = 7 [5~9] chunks
δ_{WM} = 7 [5~226] sec
δ_{WM} (1 chunk) = 73 [73~226] sec
δ_{WM} (3 chunks) = 7 [5~34] se
κ_{WM} = Acoustic or Visual
Cognitive Processor
τ_C = 70 [25~170] msec
Perceptual Processor
τ_P = 100 [50~200] msec
Motor Processor
τ_M = 70 [30~100] msec
Eye movement = 230 [70~700] msec

(4) Motor processor

Receive input from the cognitive processor and translates thoughts into actions

Execute motor programs

- Pianist: up to 16 finger movements per second

People

- Good
  1. xxx
  2. yyy
  3. zzz

- Bad
  1. aaa
  2. bbb
  3. ccc

Fill in the columns - what are people good at and what are people bad at?

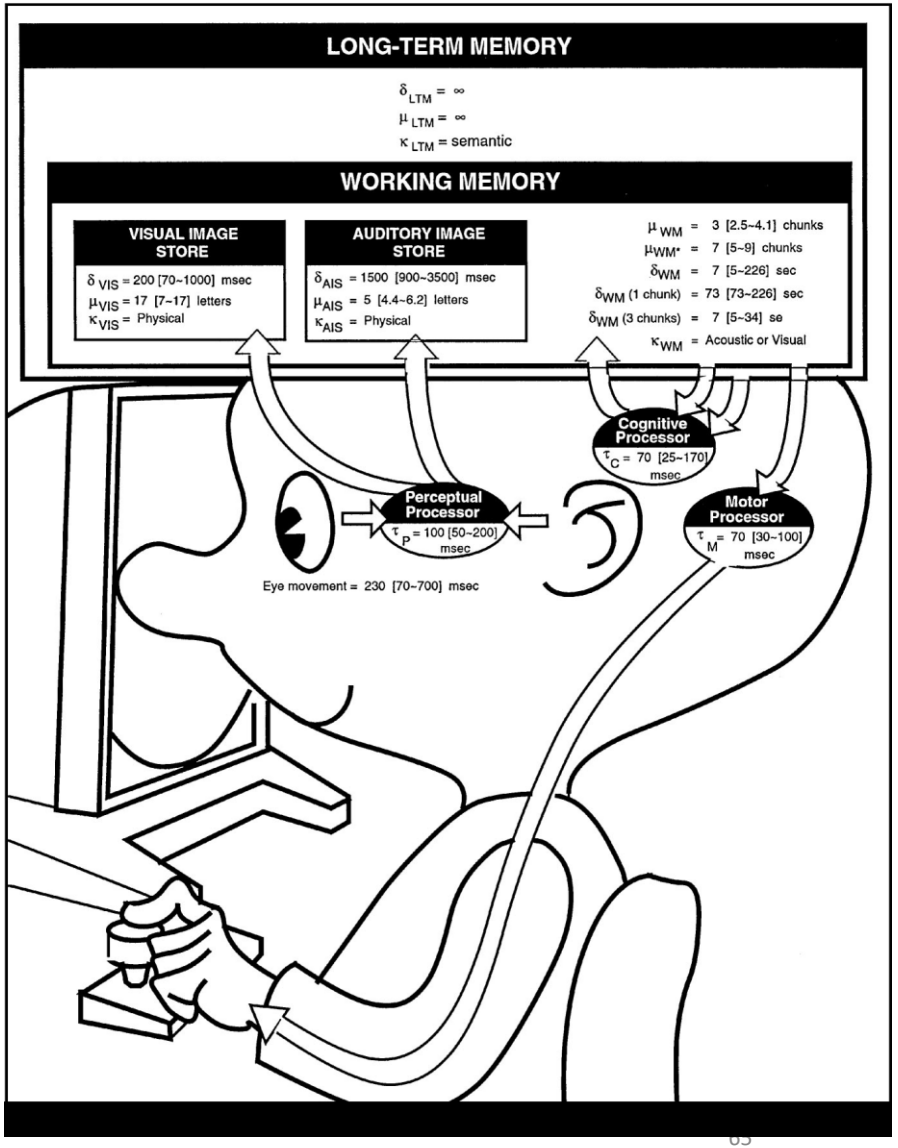
People

- Good
  - Infinite capacity LTM
  - LTM duration & complexity
  - High-learning capability
  - Powerful attention mechanism
  - Powerful pattern recognition
- Bad
  - Limited capacity STM
  - Limited duration STM
  - Unreliable access to LTM
  - Error-prone processing
  - Slow processing

Computer is opposite!
Allow one who does it best to do it!
(Function allocation)

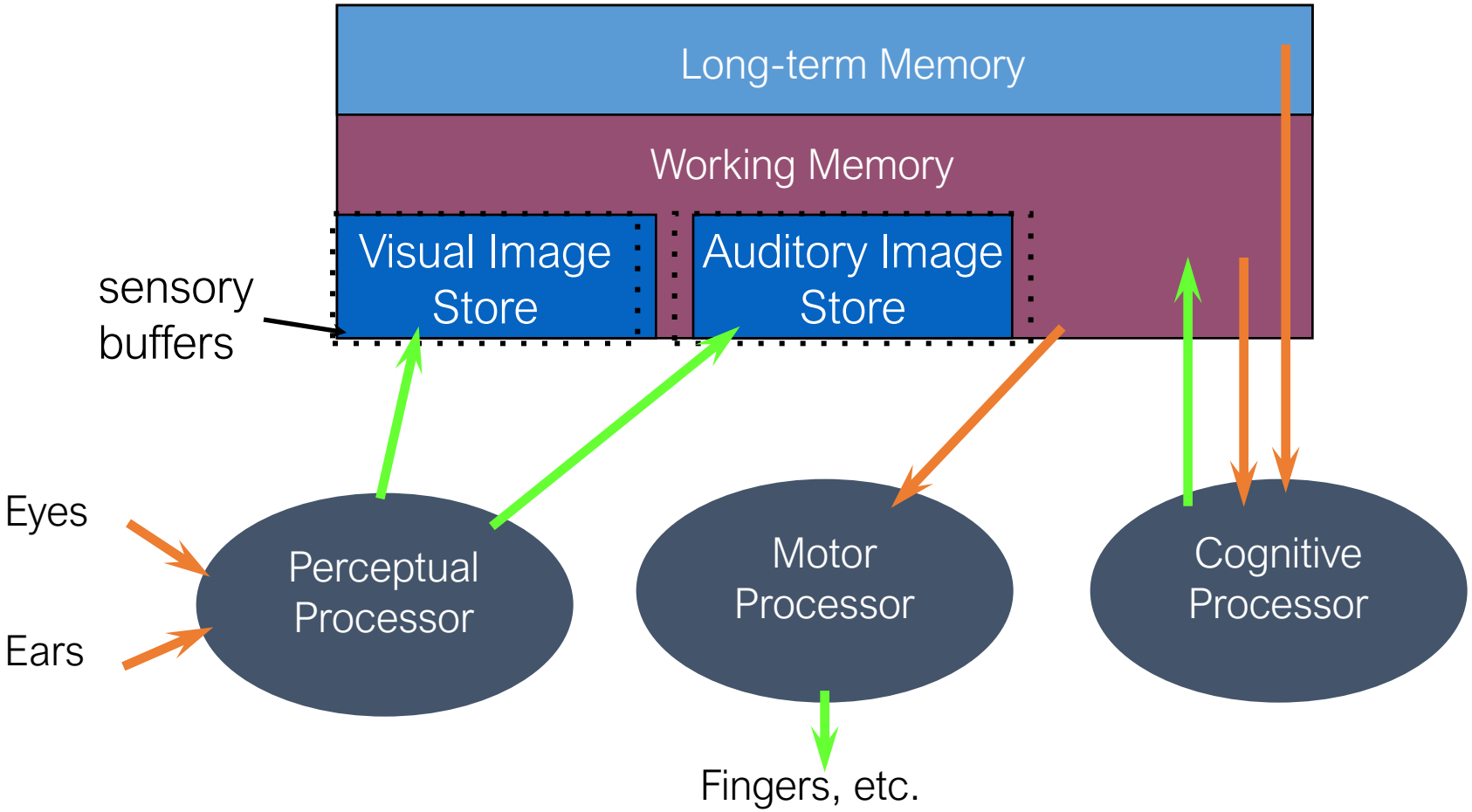
The Model Human Processor

- Developed by Card, Moran, & Newell (’83)
  - based on empirical data

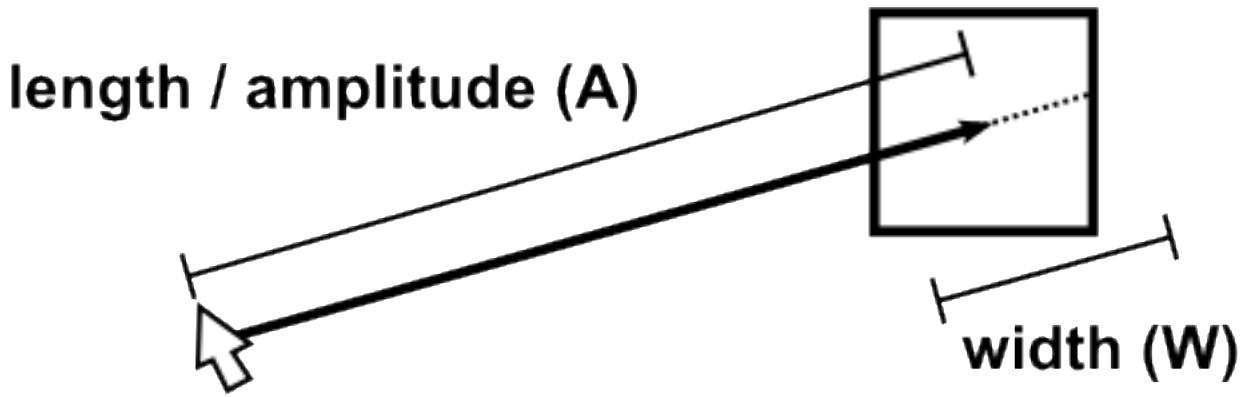


The Model Human Processor

- Developed by Card, Moran, & Newell ('83)
  - based on empirical data



Fitts's law



Fitts's law

- Fitts's law predict the amount of time taken to move to and select a target
- It is faster to hit larger targets close to you than smaller targets further from you

$T = a + b\log(D / S + 1)$

- a, b = constants (empirically derived)
- D = distance
- S = size

- T increases as the distance to the target increases
- T decreases as the size of the target increases

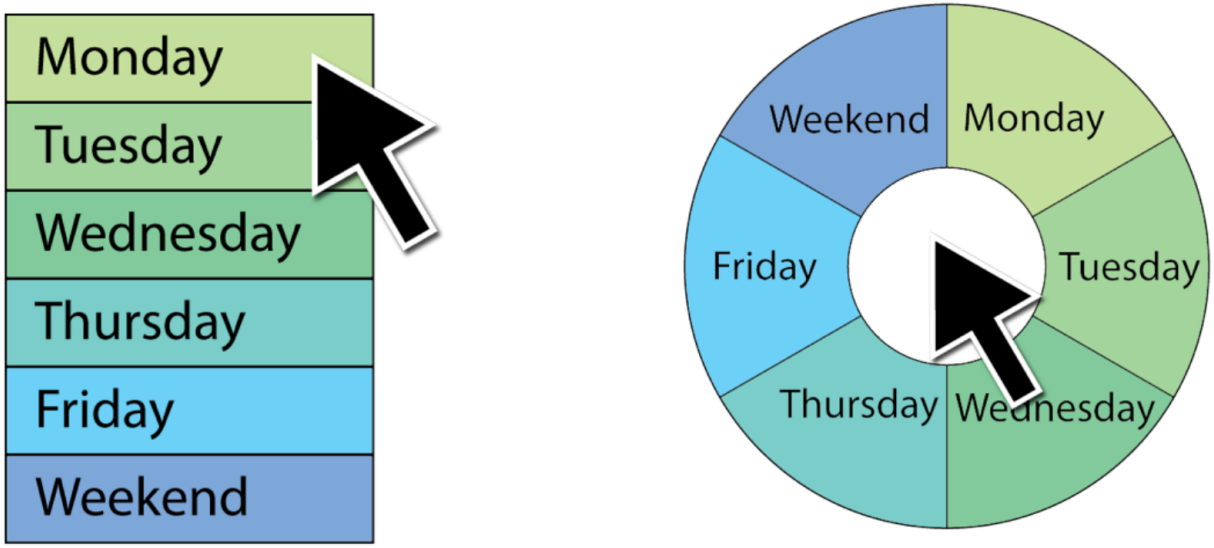
Applying Fitts's law to UI design

- Bring items closer to the cursor
- Increase the target size
- Exploit the edges

Applying Fitts's law to UI design



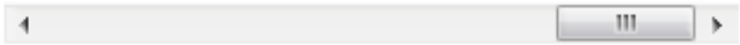
Applying Fitts's law to UI design



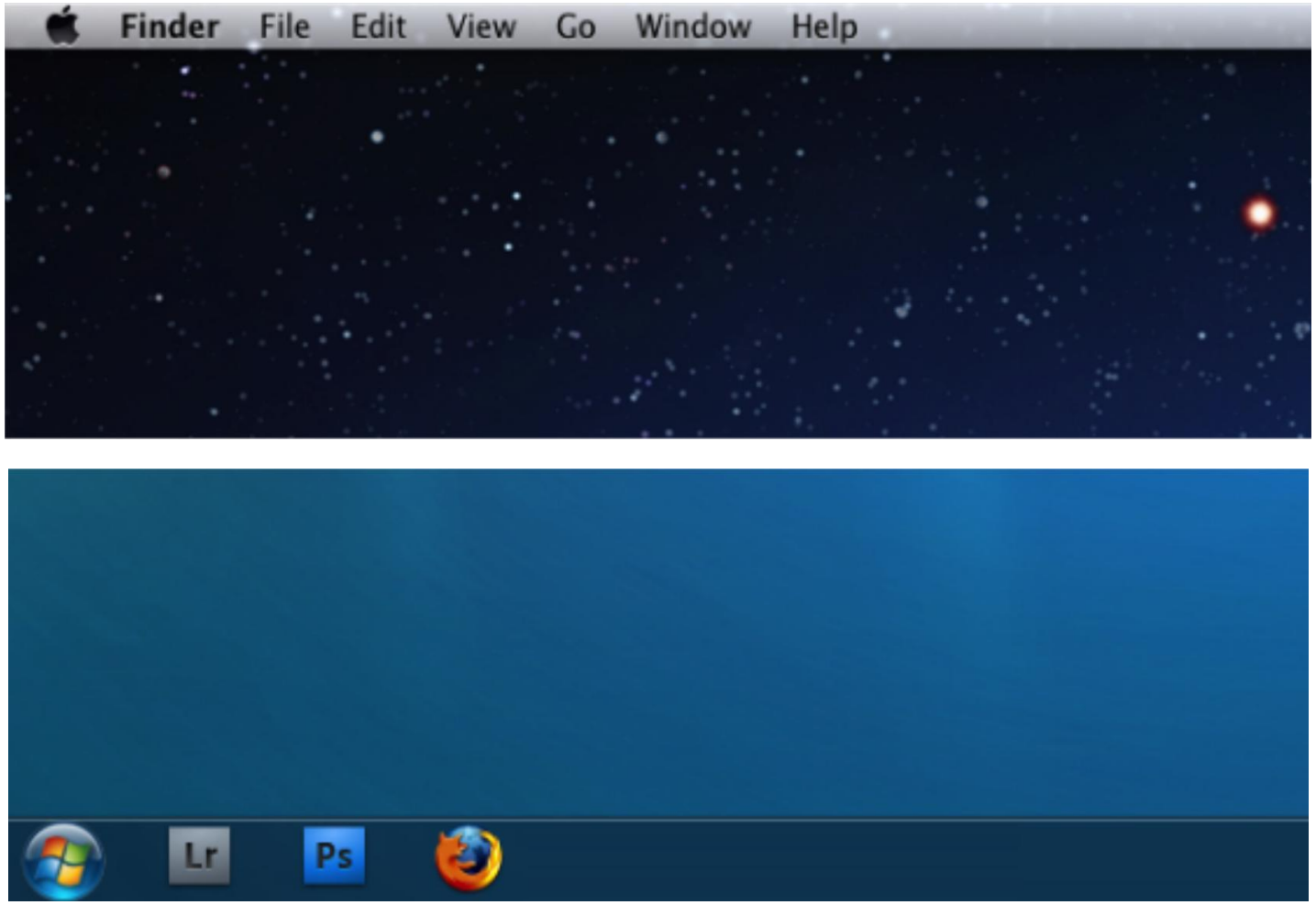
OSX Snow Leopard



Windows



Applying Fitts's law to UI design



Applying Fitts's law to UI design



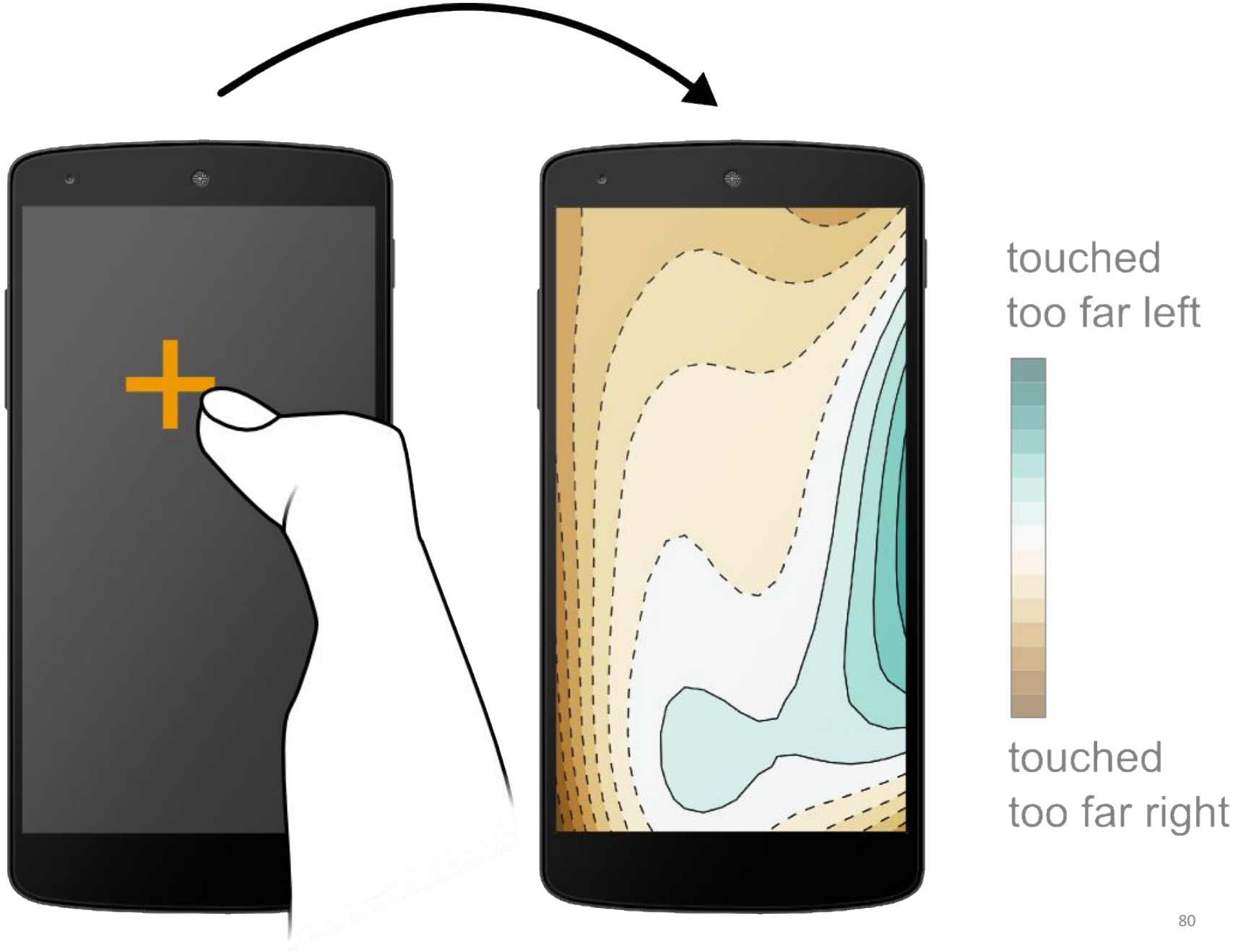
Does Fitts's law apply to mobile devices?

- Yes! Original experiment by Fitts was on human arm movement, not mouse pointing!
- Extension to target acquisition with mouse was a big result of Card et al. and not obvious.
- Tablet setting is closer to original experimental setting.
- No more benefit on device edges
- How device is held is important

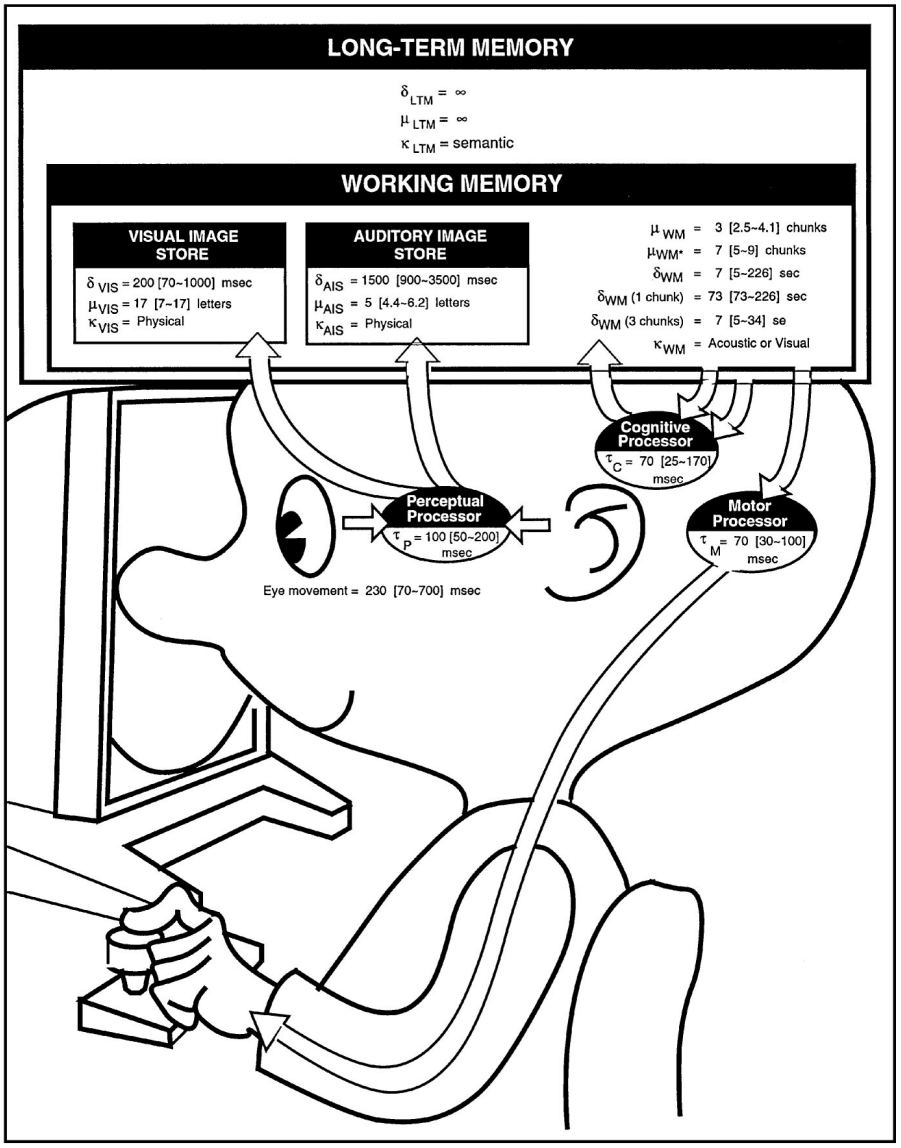

ENVATO 3G
envato


envato
ENVATO 3G

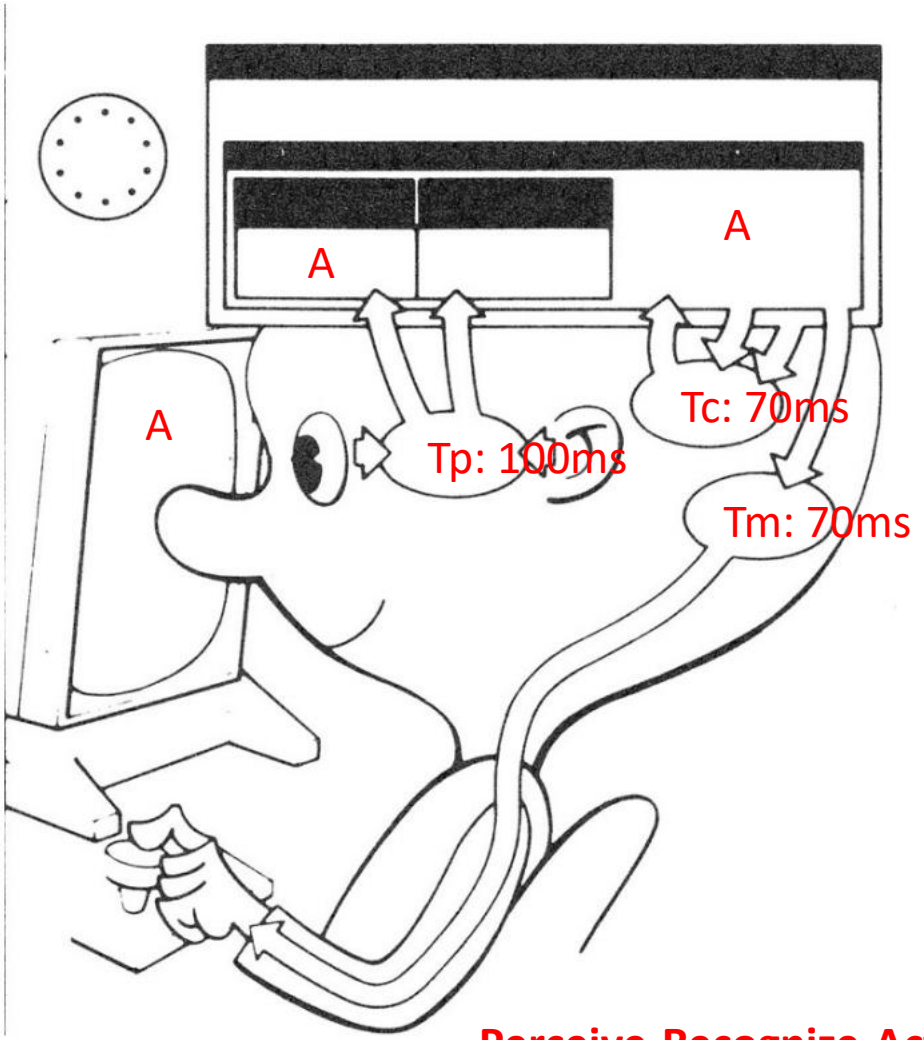

envato
ENVATO 3G


touched
too far left
touched
too far right

The Model Human Processor

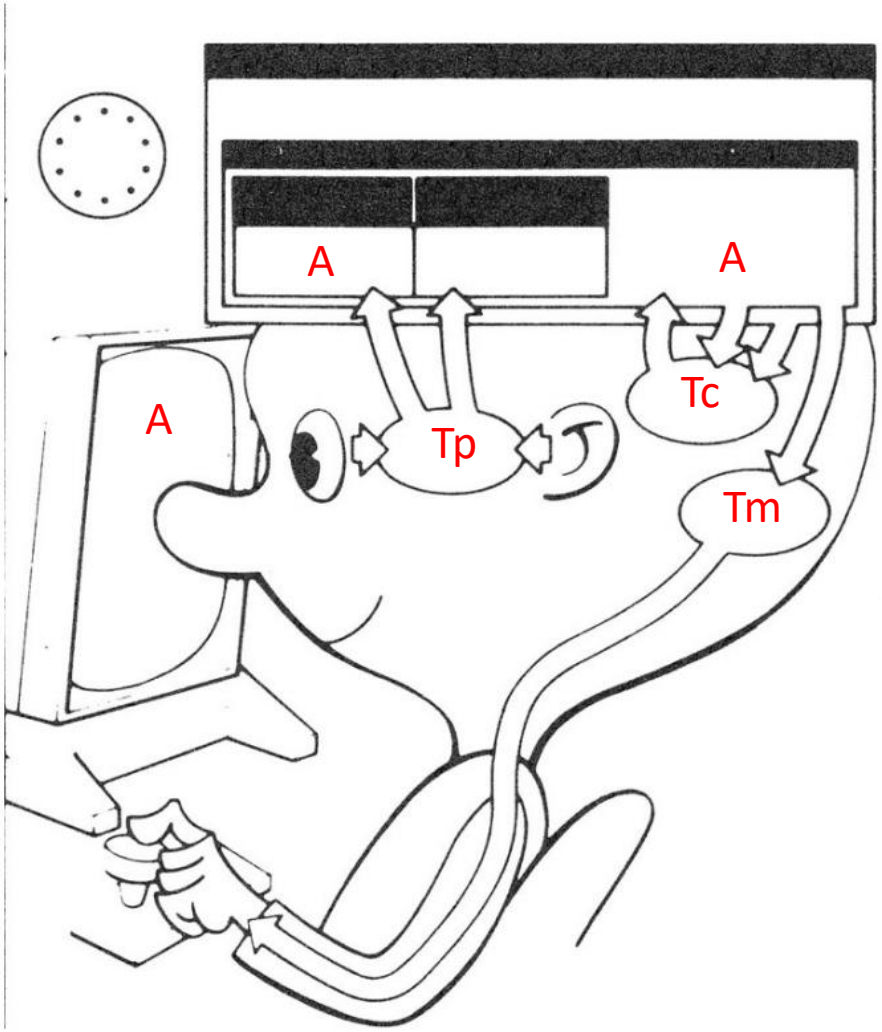


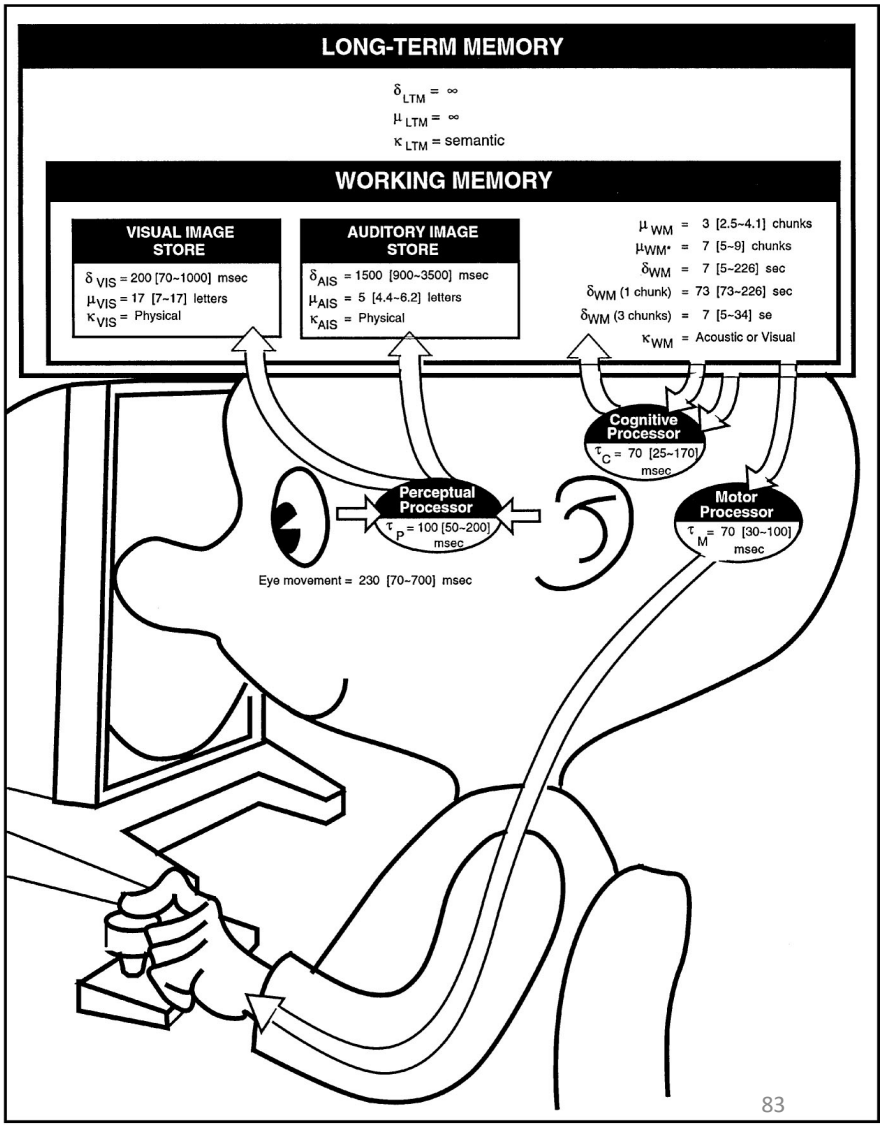
Hit Space when character appears



Perceive-Recognize-Act cycle ~= 240 ms

Hit Space when character appears





Group project

P3. Collecting Data

Collect interview data from 4 potential users (3 for TechTrio and TrafficTighten), and collect survey data from 10+ potential users. Submit the collected data.

Create a project website.

Submit:

1. Interview transcripts
2. Survey data: Raw data in a table
3. Link to the website

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

P3. Collecting Data

Format

- Add a team name and a description of your project proposal on top
- Link to your project website on top
  - Should be accessible
  - Should have following contents (no link to a pdf file):
    - Team intro
    - Project Proposal
    - Data collection plan: Interview/survey
- A PDF file, 12 point scale in Times New Roman, 1.5 line spacing
- Include: Who contributed to the original writing of each section and who helped improve and/or review at the end

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

P3. Collecting Data

Rubric

- Interview scripts (3pt)
  - -1pt per missing one interview
  - You will lose significant points if there are not enough impromptu "follow-up" questions to follow up with an interviewee's answer (e.g., what/why/how questions)
- Survey data (2pt)
- You will lose 50% if your submission does not follow the format
- You will lose 30% if it's a late submission (Not accepting a late submission after 24 hrs)

#Disclaimer. Further instruction of this submission can be given verbally during class or through Piazza.

By next class

Reading

- Contextual Design. Ch.9